DIGITAL CAKE ORDERING & MANAGEMENT SYSTEM

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Declaration of the candidate & Supervisor

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The above candidate has carried out work for the Bachelors of Information Technology Degree report under my supervision.

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Abstract

The Digital Cake Ordering & Management System is a web-based application designed to automate the cake ordering and management process for bakery businesses. The system allows customers to register and log in, browse available cake categories, view cake details, customize cakes based on size, flavor, and design, and place orders online. Customers can also view order details and track the status of their orders through the system.

From the administrative perspective, the system provides functionality to manage cake products, categories, customer details, and orders efficiently. Administrators can update order statuses, manage pricing, and maintain accurate records through a centralized management interface. The system is developed to ensure data accuracy, ease of use, and improved operational efficiency. By implementing this system, manual errors are reduced, customer satisfaction is enhanced, and bakery operations are streamlined. This project demonstrates the effective application of software engineering principles in developing a functional and user-oriented digital solution.

In addition, the system addresses key non-functional requirements such as usability, security, and reliability. It is designed with a user-friendly interface to ensure ease of use for both customers and administrators, while securely handling user data and order information. By reducing manual intervention and improving process accuracy, the system enhances customer satisfaction and supports the digital transformation of traditional bakery businesses.

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List of abbreviations

Abbreviation	Description
ACID	Atomicity, Consistency, Isolation, Durability
API / RESTful API	Application Programming Interface
AWS	Amazon Web Services
CODL	Center for Open and Distance Learning
CRUD	Create, Read, Update, Delete
CSS / CSS3	Cascading Style Sheets
DB	Database
DBMS	Database Management System
ERD	Entity Relationship Diagram
GUI	Graphical User Interface
HTML / HTML5	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
JS	JavaScript
KPI	Key Performance Indicator
MVC	Model–View–Controller
RDBMS	Relational Database Management System
SQL	Structured Query Language
SRS	Software Requirements Specification
UAT	User Acceptance Testing
UI	User Interface
UML	Unified Modeling Language
UX	User Experience

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1. Introduction

This section provides a detailed introduction to the Digital Cake Ordering & Management System, presenting essential background information required to understand the project domain. It introduces the context of online food ordering systems, outlines the purpose of the proposed system, and highlights the need for a digital solution in managing cake orders efficiently. The information presented in this section serves as preliminary data that enables the reader to gain a clear understanding of the project before proceeding to the detailed analysis and design discussed in the subsequent sections of this report.

1.1 Background

With the rapid advancement of information technology and widespread internet usage, many business activities have transitioned from traditional manual processes to digital platforms. The food industry, particularly bakery businesses, has also experienced this transformation as customers increasingly prefer online services for ordering food products. Digital ordering systems provide convenience, reduce waiting time, and improve accessibility for customers while enabling businesses to manage operations more efficiently.

Traditionally, cake ordering in bakeries is handled through physical visits or telephone calls. These methods are highly dependent on manual communication and record keeping, which can lead to order inaccuracies, delays, and difficulties in managing customer requests such as cake customization. In addition, manual systems make it challenging for bakery owners to track orders, manage inventory, and analyze customer demand effectively.

The Digital Cake Ordering & Management System is proposed to overcome these limitations by introducing an integrated web-based solution. The system allows customers to browse cake categories, view detailed product information, customize cakes according to their preferences, and place orders online. At the same time, it provides bakery administrators with tools to manage products, orders, and customer data through a centralized system. By implementing this system, bakeries can improve service quality,

reduce operational errors, and enhance customer satisfaction, highlighting the importance and relevance of this project in a modern digital business environment.

1.2 Problem Statement

Many small and medium-scale bakery businesses currently face significant challenges in managing cake orders due to the continued reliance on manual processes. Both customers and bakery administrators are affected by these inefficiencies, particularly during the cake ordering, customization, processing, and tracking stages. In such bakery environments, orders are commonly handled through in-person visits, phone calls, or handwritten records, which limits accessibility and accuracy.

For customers, the absence of a digital platform results in difficulties when browsing available cake options, customizing orders according to personal preferences, and tracking order status after placement. These challenges become more prominent during peak periods or when orders are placed outside normal business hours, as customers have limited means to obtain information or confirm order details.

From the bakery's perspective, manual order management leads to frequent issues such as incomplete or incorrect order details, delays in order processing, challenges in handling multiple orders simultaneously, and poor record maintenance. These problems occur within small and medium-scale bakery businesses where centralized digital systems are not available to support order management operations.

The root cause of these issues is the lack of a centralized and automated digital system capable of managing customer orders, customization details, and order status updates efficiently. As a result, there exists a clear gap between the current manual ordering practices and the desired state of an automated, reliable, and user-friendly solution. Addressing this gap is essential to improve operational efficiency, minimize errors, and enhance overall customer satisfaction through the implementation of a Digital Cake Ordering & Management System.

1.3 Motivation & Significance of the Project

The primary motivation for selecting the Digital Cake Ordering & Management System as the project topic is the growing reliance on digital technologies to improve business efficiency and customer experience. With the rapid expansion of e-commerce and online service platforms, customers increasingly expect convenient and flexible ways to order products. However, many bakery businesses, especially small and medium-scale enterprises, still operate using traditional manual cake ordering methods such as in-person visits, telephone calls, and handwritten order records. These practices are time-consuming, inefficient, and highly susceptible to human errors.

From the customer's perspective, manual cake ordering systems limit accessibility to product information and customization options. Customers are often unable to clearly view available cake designs, prices, sizes, and flavors before placing an order. Additionally, the absence of an order tracking mechanism creates uncertainty regarding order status, delivery time, and order confirmation. These limitations reduce customer satisfaction and may result in lost business opportunities for bakeries.

From the bakery administrator's perspective, manual order handling leads to several operational challenges, including difficulty in managing multiple orders simultaneously, maintaining accurate customer records, and updating order statuses. During peak periods, such as festive seasons or special occasions, these issues are further intensified, increasing the likelihood of incorrect orders and delayed deliveries. The lack of a centralized system also makes it difficult to analyze customer demand, manage resources effectively, and plan future business strategies.

The significance of this project lies in its ability to address these challenges by introducing a centralized, web-based solution that automates the cake ordering and management process. The proposed system enables customers to browse cake categories, customize cakes according to their preferences, and place orders online at any time. Simultaneously, it provides administrators with efficient tools to manage products, customers, and orders through a single platform. By reducing manual intervention, the system improves accuracy, saves time, and enhances overall service quality.

From an academic perspective, this project is significant as it demonstrates the practical application of software engineering concepts such as requirement analysis, system modeling, database design, and system testing. It also provides hands-on experience in developing a real-world information system that aligns with industry practices. Therefore, the Digital Cake Ordering & Management System is both practically valuable and academically meaningful, contributing to digital transformation in the bakery industry while fulfilling the learning objectives of the degree program.

1.4 Aims and Objectives

Aim

The main aim of this project is to design and develop a Digital Cake Ordering & Management System that automates the cake ordering process and improves operational efficiency in bakery businesses by providing a user-friendly, reliable, and centralized digital solution for customers and administrators.

Objectives

The specific objectives of this project are:

- To provide customers with an online platform to browse cake categories and view detailed product information
- To enable customers to customize cakes based on size, flavor, design, and other available options
- To allow customers to place cake orders online and view order details and order status
- To support bakery administrators in managing cake products, categories, and pricing efficiently
- To enable administrators to manage customer information and monitor customer orders effectively
- To improve accuracy and efficiency in order processing by reducing manual intervention

- To enhance customer satisfaction through improved accessibility, convenience, and transparency
- To ensure the system meets key non-functional requirements such as usability, security, and reliability

2. Literature Review

The literature review presents an analysis of existing online cake ordering and food ordering systems that are similar to the proposed Digital Cake Ordering & Management System. The purpose of this section is to examine current solutions, identify their strengths and weaknesses, and understand how such systems address customer and business requirements. By reviewing existing systems, this section provides a foundation for identifying gaps and limitations that justify the need for the proposed solution.

A background analysis is carried out by studying existing web-based food and cake ordering platforms, online bakery systems, and related research articles. These systems are evaluated based on factors such as user satisfaction, functionality, ease of use, customization options, order management efficiency, and overall system performance. This analysis helps to understand common features implemented in existing systems as well as recurring issues faced by users and administrators.

In addition, general opinions and feedback from various stakeholders, including customers and bakery administrators, are considered to gain insight into real-world usage and expectations. Stakeholder feedback highlights the importance of features such as easy navigation, cake customization, reliable order tracking, and efficient administrative control. The findings from this review are used to identify shortcomings in existing solutions and to guide the design of the proposed system, ensuring that it effectively addresses user needs and improves upon current practices.

2.1 Critical Evaluation of 5 Existing Solutions

This section critically evaluates five existing online cake and bakery ordering systems that are relevant to the proposed Digital Cake Ordering & Management System. The evaluation focuses on key aspects such as user satisfaction, requirement satisfaction, cake

customization capability, and administrative management features. Screenshots and comparison tables are used to highlight important similarities, differences, and limitations among the selected systems.

2.1.1 Kapruka

Kapruka is a popular Sri Lankan online ordering and delivery platform that includes a large variety of cakes for different occasions such as birthdays, anniversaries, and special celebrations. Users can browse cake categories, select designs, and order for delivery across Sri Lanka. Island-wide delivery including same-day service in Colombo and express delivery to other districts is a key feature that enhances customer convenience.

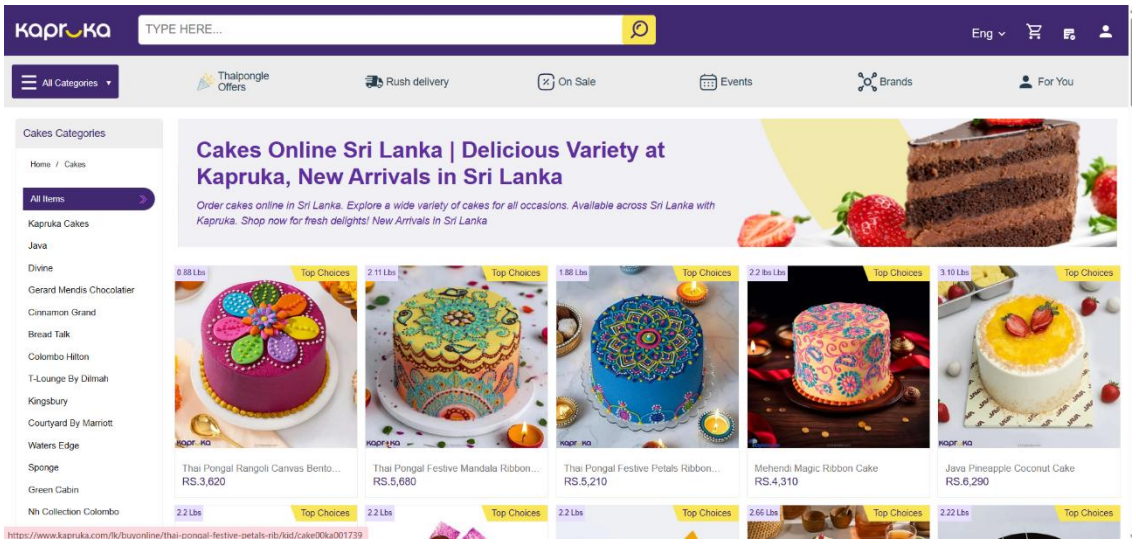

Figure 2.1: Kapruka Cake Category Interface

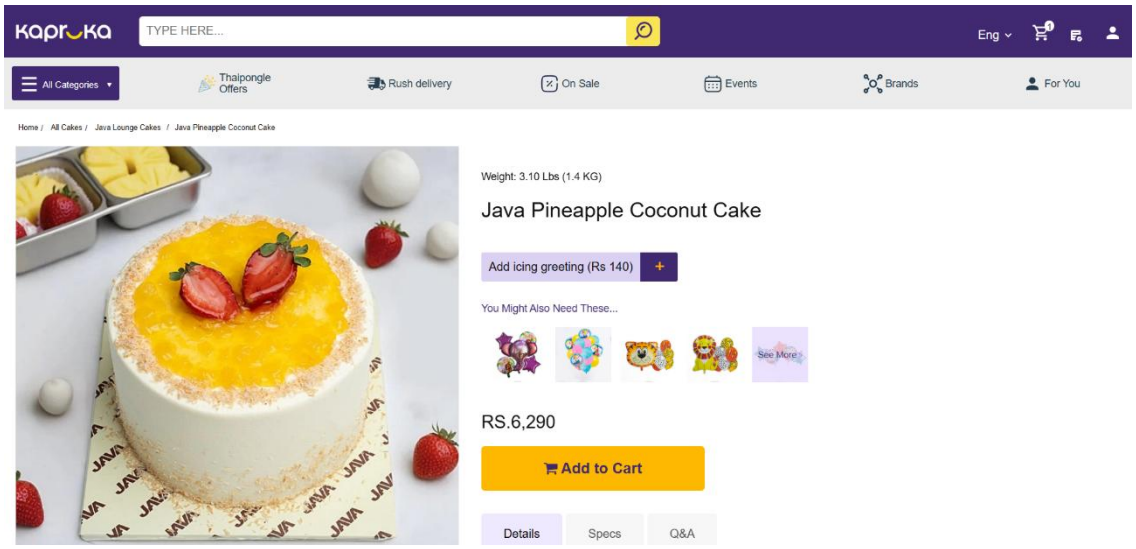

Figure 2.2: Kapruka Cake Ordering Interface

Advantages

- Provides local online cake ordering with a wide range of cake types and designs tailored to Sri Lankan customers.
- Supports multiple payment options including cash on delivery and card payments in LKR, which improves accessibility for customers who prefer local currency and payment flexibility.
- Offers same-day or next-day delivery across most regions of Sri Lanka, which is valuable for urgent orders.

Disadvantages

- Though widely used, some user reports and community feedback suggest issues with reliability and service quality on certain occasions, such as delivery challenges and payment handling indicating potential concerns about consistency.
- The platform is not specifically focused on granular cake customization workflows (e.g., rich custom topping selection and visual configuration) compared to specialized cake systems.
- Administrative control of orders by individual bakeries is limited because Kapruka operates as a centralized marketplace rather than a dedicated bakery management system.

Kapruka offers local Sri Lankan customers a platform to browse and order a wide variety of cakes online, catering to birthdays, anniversaries, and celebrations. The system supports island-wide delivery and multiple payment methods, including cash on delivery and local currency payment options, which improve accessibility for users across Sri Lanka. However, user reports indicate occasional challenges with service reliability and delivery performance. Additionally, Kapruka emphasizes product selection and delivery rather than detailed customization workflows or administrative management features, which differentiates it from systems designed specifically for bakery order processing and management.

2.1.2 Taplow.lk

Taplow.lk is a Sri Lankan online bakery platform that offers a diverse range of cakes including celebration cakes, custom cakes, jar cakes, and specialty dessert items. The platform provides online browsing of cake products and supports delivery across Sri Lanka, with options such as free delivery within Colombo and scheduled delivery outside the city. Taplow's offerings include classic birthday cakes, custom-designed cakes, and creative dessert selections catering to various occasions.

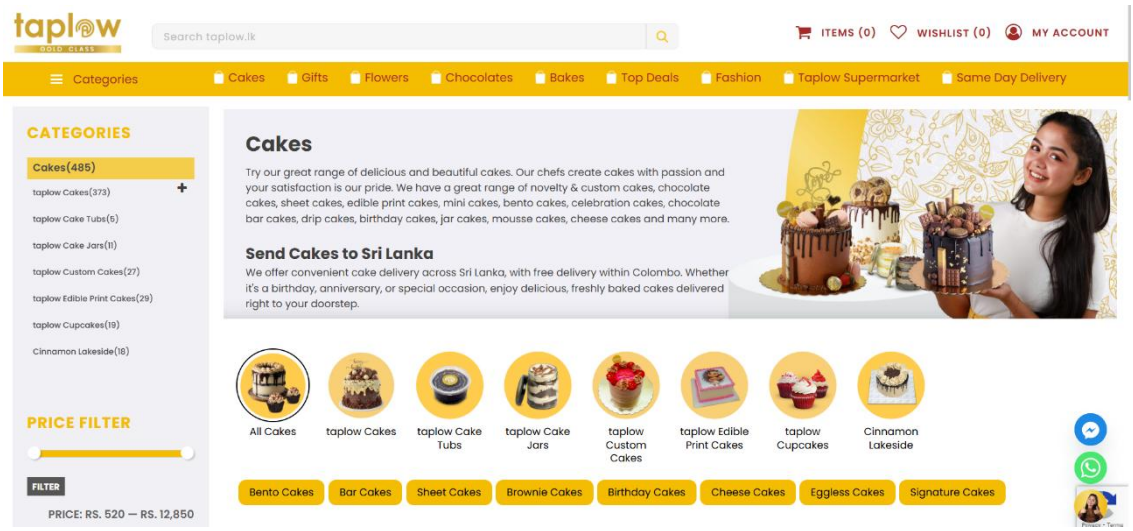

Figure 2.3: Taplow Interface

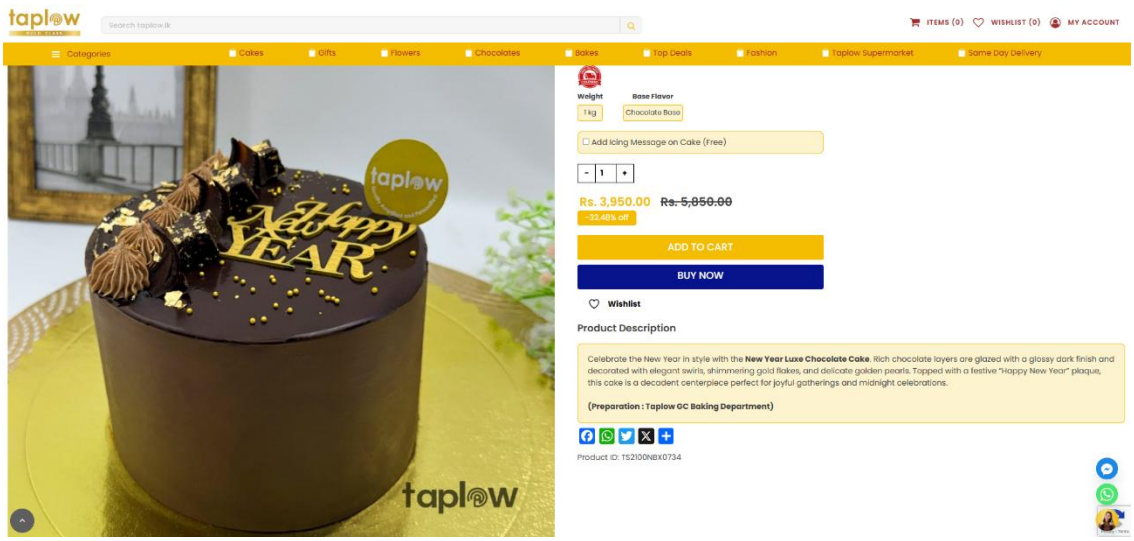

Figure 2.4: Taplow Cake Ordering Interface

Advantages

- Local Sri Lankan service with delivery across Sri Lanka, improving accessibility for domestic users.
- Wide variety of cake types including custom cakes, jar cakes, and specialty dessert products, offering more choices.
- Same-day delivery available (depending on time and location), which enhances convenience for urgent orders.
- Free delivery within Colombo and suburbs helps reduce cost for local customers.
- Website presents cake categories clearly, making it easier for users to browse and select products.

Disadvantages

- Customization options are limited compared to dedicated custom cake workflows in specialized systems no rich visual customization builder.
- Limited online order tracking is provided (focus is mainly on order placement and delivery scheduling).
- Backend administrative features for bakery management (e.g., centralized order dashboard) are not visible to individual partner bakeries and may be minimal.
- The platform has a mixed generic e-commerce feel rather than a specialized bakery management system.

Taplow.lk is a Sri Lankan online bakery platform offering a variety of cakes and desserts with island-wide delivery and free delivery within Colombo. The platform includes celebration cakes, custom cakes, and specialty desserts, catering to local customer needs. However, it provides limited customization features and lacks advanced administrative capabilities compared to dedicated bakery ordering systems.

2.1.3 Dinevate – Bakery Website with Ordering

Dinevate is a professional bakery e-commerce solution that focuses on helping bakery businesses establish an online presence through visually appealing websites and integrated online ordering features. The platform combines product display, basic ordering functionality, and digital marketing tools to support online sales and customer

engagement. It is primarily designed to enhance brand visibility and attract customers through optimized web design and marketing strategies.

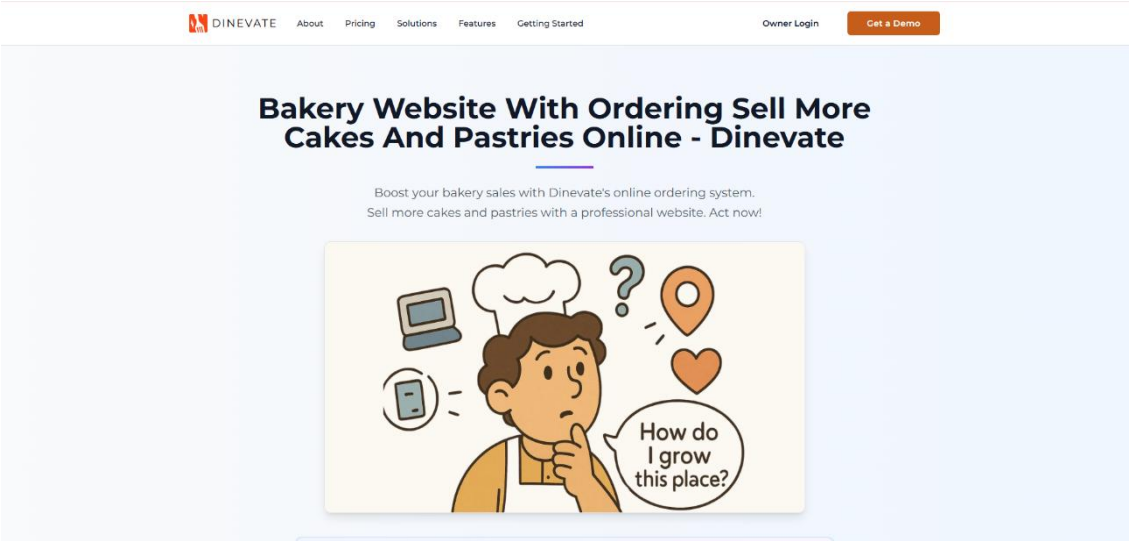

Figure 2.5: Online cake product display and ordering page of Dinevate system

Advantages

- Provides an integrated bakery e-commerce platform with online ordering capabilities
- Includes built-in SEO and digital marketing automation tools to improve online visibility
- Mobile-friendly and visually attractive interface enhances user experience and customer engagement
- Supports promotional content and online branding for bakery businesses
- Backend management tools assist in maintaining product listings and managing online content
- Suitable for bakeries aiming to strengthen their online presence and customer reach

Disadvantages

- Places greater emphasis on marketing and website presentation than on detailed order management
- Limited support for advanced cake customization and complex order handling
- Real-time order tracking and detailed order status updates are not strongly supported
- Custom pricing and service-based model may not be affordable for small-scale bakeries
- Administrative features may be insufficient for handling high volumes of orders efficiently

2.1.4 Winni – Online Cake & Gift Delivery Platform

Winni is an India-based online cake and gift delivery platform that enables customers to browse cake categories, customize products, and place orders based on delivery location and time. The system integrates product search, cake customization, cart management, and location-based delivery charge calculation. Winni primarily operates as a centralized e-commerce marketplace rather than a dedicated bakery management system.

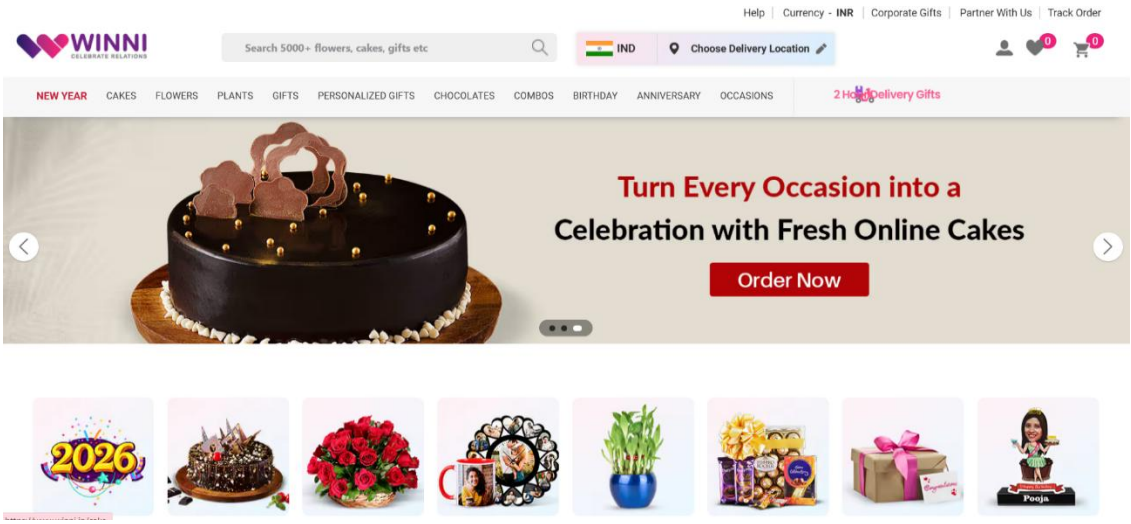

Figure 2.6: Winni landing page

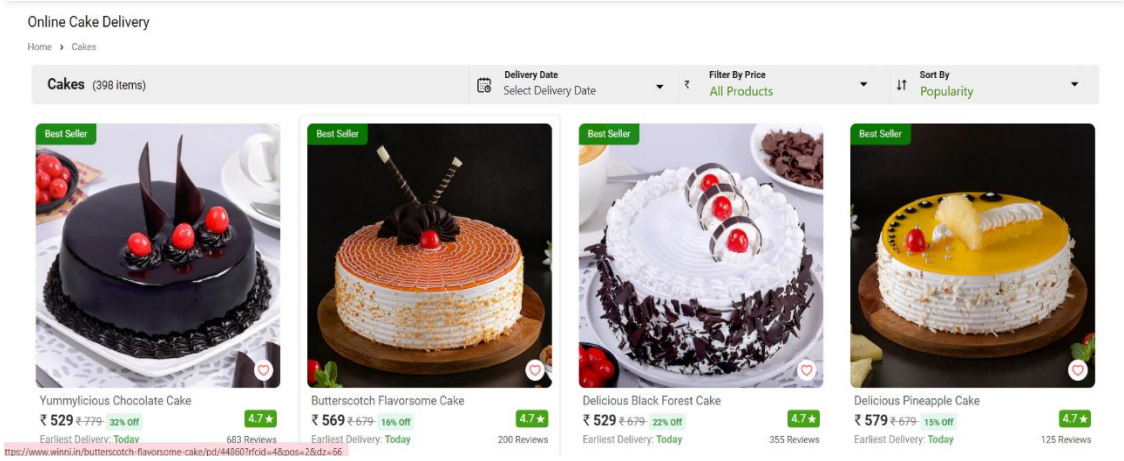

Figure 2.7: Winni product search interface showing cake categories

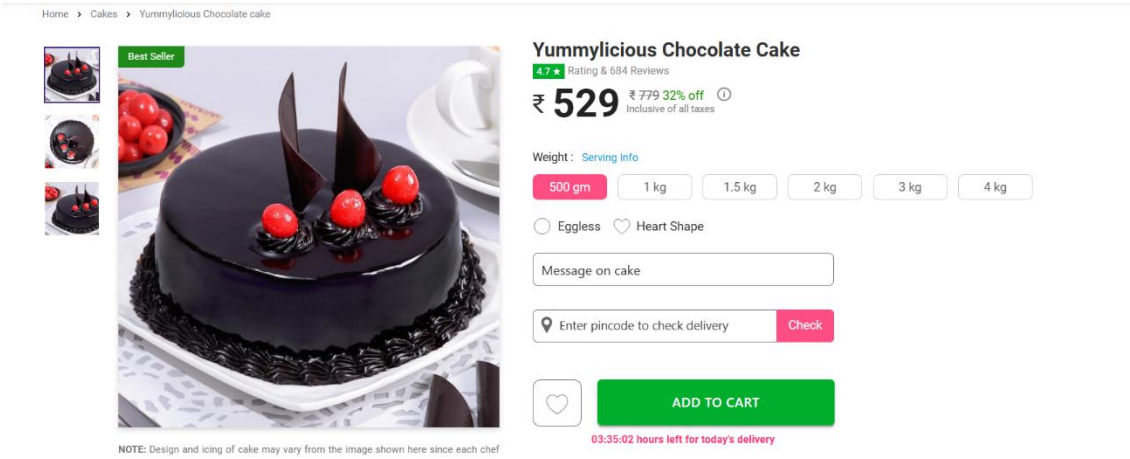

Figure 2.8: Cake customization and add-to-cart screen in Winni

Advantages

- Provides a wide range of cake categories with advanced product search functionality
- Supports basic cake customization options such as weight, flavor, and add-ons
- Location-based delivery selection improves delivery accuracy and transparency
- User-friendly interface enhances customer satisfaction
- Integrated payment gateway ensures secure online transactions
- High availability and scalability as a large commercial platform

Disadvantages

- Designed as a marketplace, not tailored for individual bakery management needs
- Limited administrative control for individual bakeries over orders and inventory
- Customization options are restricted compared to dedicated custom cake systems
- Backend order management processes are not visible or configurable for vendors
- Focuses more on customer convenience than internal production management

2.1.5 Ferns N Petals (FNP) – Cake & Gifts Ordering System

Ferns N Petals (FNP) is a well-established e-commerce platform specializing in cakes, gifts, and floral products. The system offers an advanced and visually rich user interface that allows customers to browse categories, view detailed product descriptions, select delivery dates and time slots, and place orders online. FNP emphasizes customer experience and logistics efficiency through its nationwide delivery network.

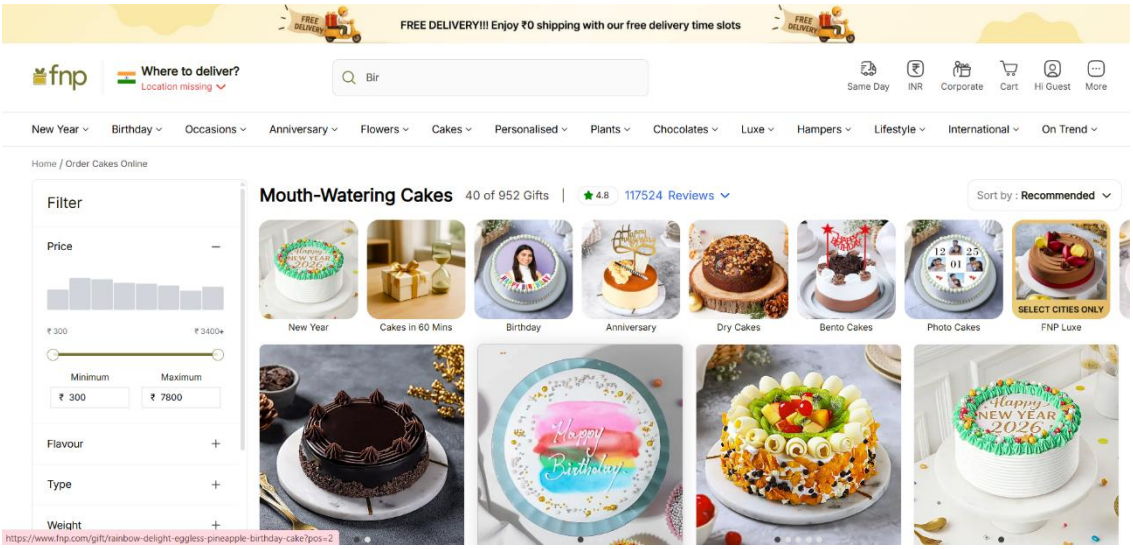

Figure 2.9: Ferns N Petals cake category listing page

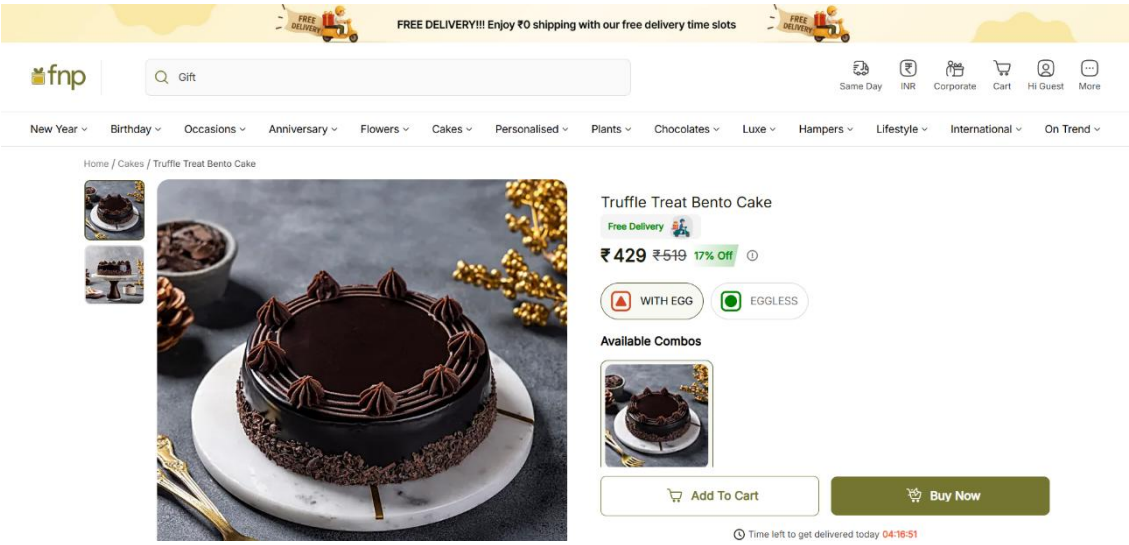

Figure 2.10: Ferns N Petals Cake product detail page with pricing and description

Advantages

- Rich and visually appealing user interface improve user engagement
- Detailed cake product pages with images, descriptions, and pricing
- Flexible delivery date and time slot selection enhances customer convenience
- Strong logistics and order tracking capabilities
- High level of customer trust due to brand recognition
- Scalable platform capable of handling large volumes of orders

Disadvantages

- Not designed as a bakery management or production planning system
- Limited cake customization compared to specialized custom cake platforms
- Backend administrative features are not accessible to individual bakery operators
- Focuses on mass-market ordering rather than personalized bakery workflows
- Integration with internal bakery operations is minimal

2.2 Comparison of Existing Online Cake Ordering Systems

Table 2.1: Comparative analysis of existing online cake ordering and bakery management systems

System	Domain Focus	Cake Customization	Order Management	Admin Dashboard	Order Tracking	User Experience	Cost / Accessibility
Kapruka	Sri Lankan cake & gift e-commerce marketplace	Limited (basic size/message options)	Basic order handling	Limited (centralized platform control)	Partial tracking	Good for local users	Moderate cost; high local accessibility
Taplow.lk	Sri Lankan online bakery platform	Limited (manual customization requests)	Basic order processing	Not available	Minimal	Simple and clear UI	Affordable; high accessibility within Sri Lanka
Dinevate	Bakery e-commerce with marketing focus	Basic customization	Moderate order handling	Available (limited bakery control)	Limited	High visual appeal	Subscription-based; moderate accessibility
Winni	Online cake & gift ordering marketplace	Moderate (flavor, message, add-ons)	Moderate order processing	Not available for vendors	Good tracking	User-friendly and responsive	Moderate pricing; region-based access
Ferns N Petals (FNP)	Large-scale cake & gift e-commerce platform	Moderate to high	Advanced order management	Centralized platform control	Advanced tracking	High-quality user experience	Higher cost; wide regional access

2.3 Narrative Evaluation of Existing Systems

The evaluation of existing cake and bakery ordering systems reveals that while several platforms successfully support online ordering and delivery, none fully satisfy the combined requirements of rich cake customization, efficient order management, and dedicated administrative control required by small and medium-scale bakery businesses.

Sri Lankan platforms such as Kapruka and Taplow.lk provide strong local accessibility by supporting local currency payments and island-wide delivery. These systems are effective in enabling customers to place orders online; however, they primarily function as e-commerce marketplaces rather than comprehensive bakery management systems. Cake customization features are limited, often relying on manual communication, and administrative dashboards for bakery owners are either restricted or unavailable. As a result, these platforms offer convenience but lack operational flexibility and internal management capabilities.

International platforms such as Winni and Ferns N Petals (FNP) demonstrate advanced e-commerce capabilities, including improved user experience, product categorization, and delivery tracking. FNP, in particular, offers robust logistics and order tracking features suitable for high-volume operations. However, both systems operate under centralized control models, limiting direct administrative access for individual bakeries. Customization options, while better than local platforms, are still constrained and not designed to support detailed, step-by-step cake configuration workflows.

Dinevate differs from the other systems by placing a strong emphasis on digital marketing and visual presentation. Its user interface is highly appealing and supports basic ordering functionality. Nevertheless, its focus on marketing automation outweighs its support for operational order management, resulting in limited customization capabilities and minimal administrative control for bakery-specific processes.

Overall, the analysis indicates that existing systems either prioritize customer-facing e-commerce features or centralized marketplace operations, while neglecting the need for a unified platform that supports both customers and bakery administrators effectively. The absence of an integrated solution that combines advanced cake customization, centralized order management, real-time order tracking, and a dedicated administrative

dashboard highlights a clear functional gap. This gap directly supports the necessity for the proposed Digital Cake Ordering & Management System, which aims to address these limitations and provide a tailored solution for small and medium-scale bakery businesses.

2.4 Background analysis and a survey

Introduction to the Survey

A survey was conducted to collect user opinions and requirements related to online cake ordering systems. The main objective of this survey was to identify user satisfaction levels, existing problems, and expected features in current cake ordering platforms. The survey results were used to support the requirement analysis and to justify the need for the proposed Digital Cake Ordering & Management System.

The questionnaire was distributed using Google Forms and responses were collected from users who have prior experience with ordering cakes online.

Survey Methodology

- **Survey tool:** Google Forms
- **Target participants:**
    - Customers who order cakes online
    - Individuals familiar with online food or bakery ordering systems
- **Type of questions:**
    - Multiple choice questions
    - Opinion-based questions
- **Focus areas:**
    - Ease of ordering cakes online
    - Cake customization features
    - Order tracking and delivery transparency

Survey Question 1: Age Group of the Respondents

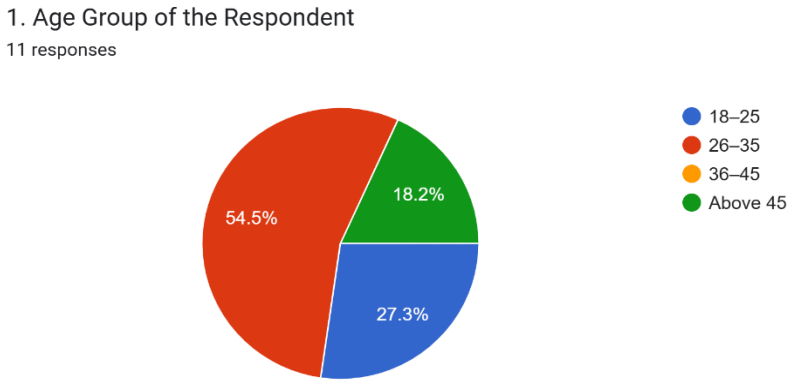

Figure 2.11: Survey Q1

The survey received a total of 11 responses, and the age distribution of respondents is presented in the table above. The majority of respondents (54.5%) belong to the 26–35 age group, indicating that this demographic represents the primary user base for the Digital Cake Ordering and Management System. Users in this age range are typically familiar with online platforms and are more likely to adopt digital ordering systems.

A significant proportion of respondents (27.3%) fall within the 18–25 age group, further supporting the need for a modern, intuitive, and mobile-friendly user interface. Additionally, 18.2% of respondents are above 45 years of age, highlighting the importance of accessibility features such as clear navigation, readable fonts, and simple workflows to accommodate users with varying levels of technical experience.

The absence of respondents in the 36–45 age group does not negatively impact the system design, as the identified user groups still represent a broad range of potential customers. Overall, the results justify the requirement for a responsive, user-friendly, and accessible web-based system suitable for diverse age groups.

Table 2.2: Survey Q1 Analysis

Age Group	Number of Responses	Percentage (%)
18–25	3	27.3%
26–35	6	54.5%
36–45	0	0.0%
Above 45	2	18.2%
Total	**11**	**100%**

Survey Question 2: Do you own a cake business?

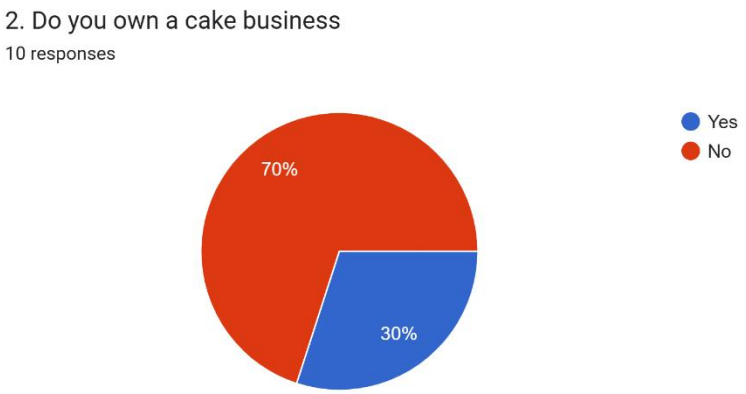

Figure 2.12: Survey Q2

A total of 10 responses were collected for this question. As shown in the table above, 70% of respondents do not own a cake business, while 30% indicated that they own a cake business. This result confirms that the majority of survey participants are customers rather than business owners, validating the system's primary focus on customer-oriented functionalities such as cake browsing, customization, online order placement, and order tracking.

At the same time, the presence of 30% business owners demonstrates the importance of including administrative features within the system, such as order management, inventory management, and product management. This balanced representation supports the decision to design the system with two distinct user roles: Customer and

Administrator, each with clearly defined access privileges and functionalities.

Table 2.3: Survey Q2 Analysis

Response	Number of Responses	Percentage (%)
Yes	3	30%
No	7	70%
Total	**10**	**100%**

Survey Question 3: Frequency of Cake Ordering

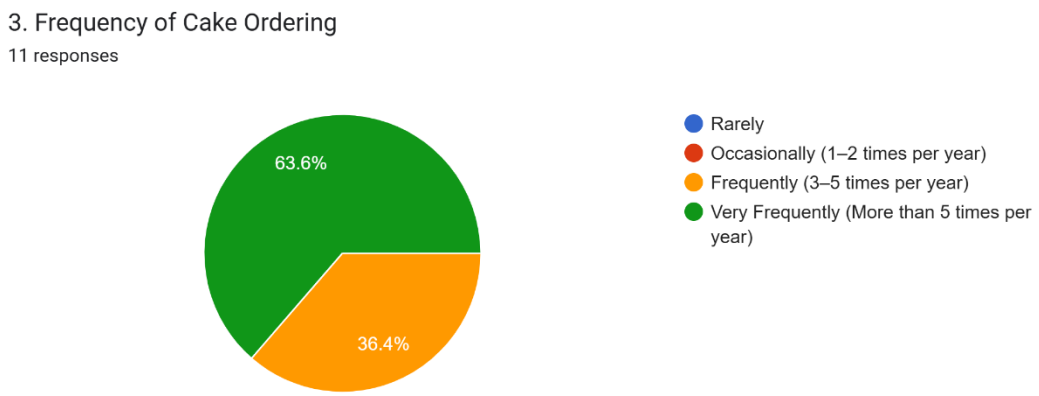

Figure 2.13: Survey Q3

The survey collected 11 responses regarding the frequency of cake ordering. As shown in the table above, a significant majority of respondents (63.6%) order cakes very frequently, defined as more than five times per year. Additionally, 36.4% of respondents indicated that they order cakes frequently, within the range of three to five times per year.

Notably, there were no respondents who selected the *Rarely* or Occasionally options. This indicates that the survey participants are regular cake buyers, making them a highly relevant user group for evaluating the requirements of an online cake ordering system.

These results strongly justify the need for an efficient, reliable, and user-friendly digital ordering platform that supports repeated usage. Features such as quick order placement,

saved preferences, order tracking, and streamlined workflows become essential when serving frequent users.

Table 2.4: Survey Q3 Analysis

Frequency Category	Number of Responses	Percentage (%)
Rarely	0	0.0%
Occasionally (1–2 times per year)	0	0.0%
Frequently (3–5 times per year)	4	36.4%
Very Frequently (More than 5 times per year)	7	63.6%
Total	**11**	**100%**

Survey Question 4: Previous Experience with Online Cake Ordering

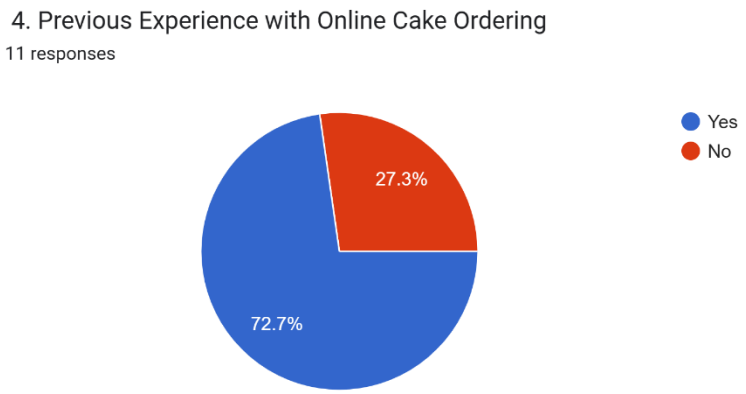

Figure 2.14: Survey Q4

A total of 11 responses were recorded for this question. The results show that 72.7% of respondents have previous experience with online cake ordering, while 27.3% have not used an online cake ordering system before. This indicates that the majority of users are already familiar with digital ordering platforms, which supports the feasibility and acceptance of an online-based solution.

However, the presence of users without prior experience highlights the importance of designing the system with a simple, intuitive interface and clear guidance to ensure ease of use for first-time users. Features such as clear navigation, straightforward order steps,

and user-friendly instructions are essential to accommodate users with varying levels of technical familiarity.

Overall, these results validate the requirement for a user-centered design approach that balances functionality for experienced users while maintaining accessibility for new users.

Table 2.5: Survey Q4 Analysis

Response	Number of Responses	Percentage (%)
Yes	8	72.7%
No	3	27.3%
Total	**11**	**100%**

Survey Question 5: Challenges Faced in Traditional Cake Ordering

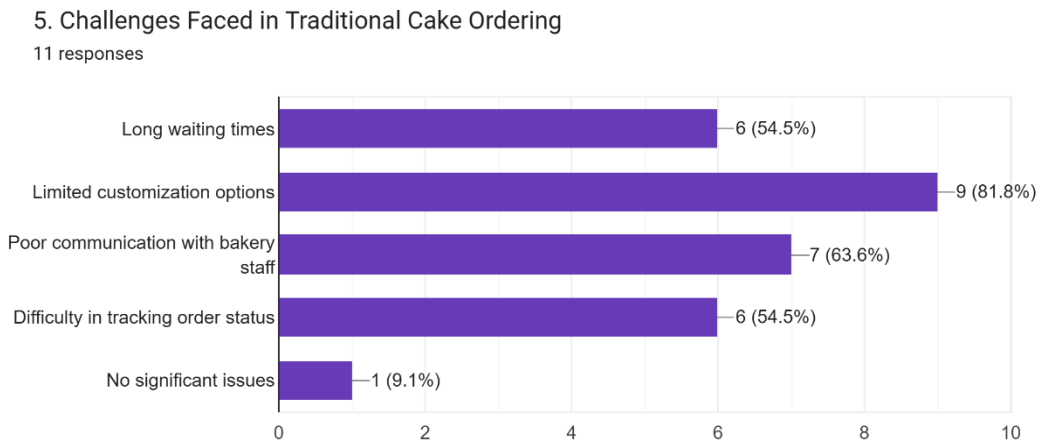

Figure 2.15: Survey Q5

A total of 11 responses were collected for this question, with respondents allowed to select multiple challenges. The results indicate that limited customization options are the most significant issue, reported by 81.8% of respondents. This highlights a strong user demand for flexible and detailed cake customization features within the proposed system.

Additionally, 63.6% of respondents identified poor communication with bakery staff as a major challenge, while 54.5% reported long waiting times and difficulty in tracking order status. These findings demonstrate inefficiencies in traditional ordering processes and emphasize the need for improved communication and transparency.

Only 9.1% of respondents indicated that they experienced no significant issues, suggesting that the majority of users face at least one problem when ordering cakes through traditional methods. Overall, these results strongly justify the development of a Digital Cake Ordering and Management System to address existing limitations and enhance customer experience.

Table 2.6: Survey Q5 Analysis

Challenge	Number of Responses	Percentage (%)
Long waiting times	6	54.5%
Limited customization options	9	81.8%
Poor communication with bakery staff	7	63.6%
Difficulty in tracking order status	6	54.5%
No significant issues	1	9.1%

Survey Question 6: Preference for an Online Cake Ordering System

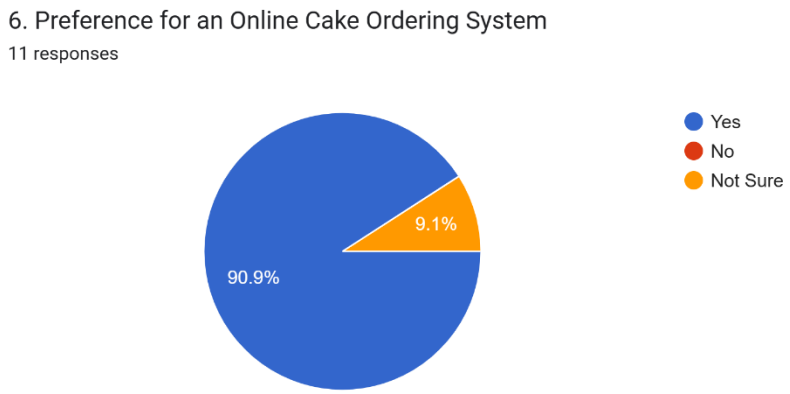

Figure 2.16: Survey Q6

A total of 11 responses were collected for this question. The results clearly indicate a strong preference for an online cake ordering solution, with 90.9% of respondents expressing interest in using an online system. Notably, no respondents selected the No option, demonstrating a high level of acceptance and demand for digital ordering platforms.

Only 9.1% of respondents were Not Sure, suggesting minor uncertainty that can be addressed through effective system design, usability, and user education. Overall, this overwhelming positive response strongly validates the necessity and relevance of developing the Digital Cake Ordering and Management System.

These findings confirm that an online solution would be well-received by users and support the system's primary objective of improving convenience, efficiency, and user satisfaction.

Table 2.7: Survey Q6 Analysis

Response	Number of Responses	Percentage (%)
Yes	10	90.9%
No	0	0.0%
Not Sure	1	9.1%
Total	**11**	**100%**

Survey Question 7: Preferred Cake Customization Options

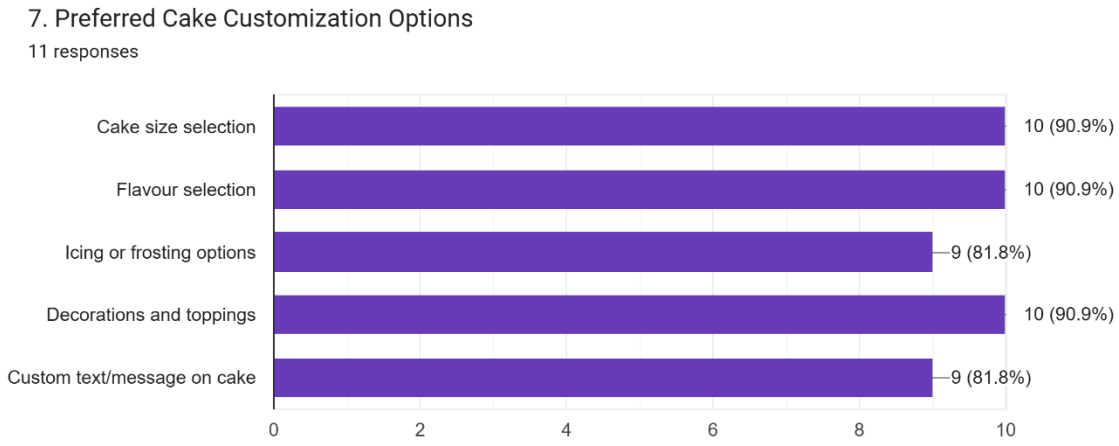

Figure 2.17: Survey Q7

A total of 11 responses were collected for this question, with respondents allowed to select multiple options. The results show a strong preference for comprehensive cake customization features. Cake size selection, flavor selection, and decorations and toppings were each selected by 90.9% of respondents, indicating that these features are considered essential by most users.

Additionally, icing or frosting options and custom text or message on the cake were selected by 81.8% of respondents, further emphasizing the importance of personalized cake design. These findings demonstrate that users expect a high level of flexibility and personalization when ordering cakes online.

Overall, the results strongly justify the inclusion of an advanced Cake Customization Module that allows users to tailor cakes according to their individual preferences. Such features are critical for improving user satisfaction and aligning the system with real-world customer expectations.

Table 2.8: Survey Q7 Analysis

Customization Option	Number of Responses	Percentage (%)
Cake size selection	10	90.9%
Flavor selection	10	90.9%
Icing or frosting options	9	81.8%
Decorations and toppings	10	90.9%
Custom text/message on cake	9	81.8%

Survey Question 8: Preferred Order Fulfilment Method

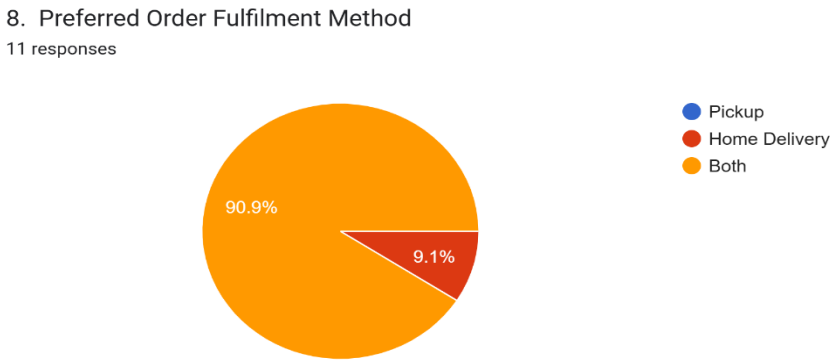

Figure 2.18:Survey Q8

A total of 11 responses were recorded for this question. The results indicate that 90.9% of respondents prefer having both pickup and home delivery options available, demonstrating a strong demand for flexibility in order fulfilment. This suggests that users value the ability to choose their preferred method based on convenience, urgency, or personal circumstances.

Only 9.1% of respondents preferred home delivery only, while no respondents selected the pickup only option. This further emphasizes the importance of supporting multiple fulfilment methods rather than limiting the system to a single option.

Overall, these findings strongly justify the inclusion of both pickup scheduling and home delivery management within the system to meet diverse user needs and enhance overall user satisfaction.

Table 2.9: Survey Q8 Analysis

Fulfilment Method	Number of Responses	Percentage (%)
Pickup only	0	0.0%
Home delivery only	1	9.1%
Both pickup and delivery	10	90.9%
Total	**11**	**100%**

Survey Question 9: Importance of Order Tracking Functionality

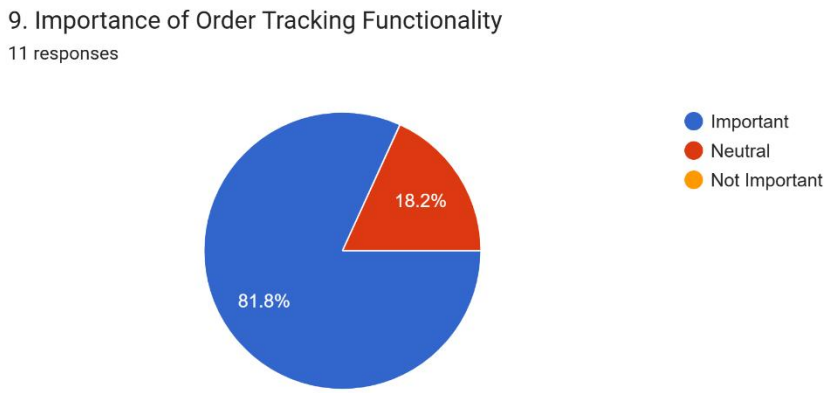

Figure 2.19: Survey Q9

A total of 11 responses were collected regarding the importance of order tracking functionality. The results show that 81.8% of respondents consider order tracking to be important, indicating a strong expectation for visibility into order status after placement. This reflects user demand for transparency and reassurance throughout the ordering process.

The remaining 18.2% of respondents selected *Neutral*, suggesting that while order tracking may not be critical for all users, it is still perceived as a valuable feature. Notably, no respondents indicated that order tracking is not important, further reinforcing its relevance.

Overall, these findings strongly justify the inclusion of an Order Tracking Module within the system to enhance user confidence, reduce uncertainty, and improve overall customer experience.

Table 2.10: Survey Q9 Analysis

Importance Level	Number of Responses	Percentage (%)
Important	**9**	**81.8%**
Neutral	**2**	**18.2%**
Not Important	**0**	**0.0%**
Total	**11**	**100%**

Administrator Requirements (Optional)

This section is intended for bakery owners or staff members to identify challenges and requirements related to administrative tasks.
The responses help determine the need for system features such as order management, inventory control, and administrative dashboards.

Survey Question 10: Difficulty in Managing Orders Manually

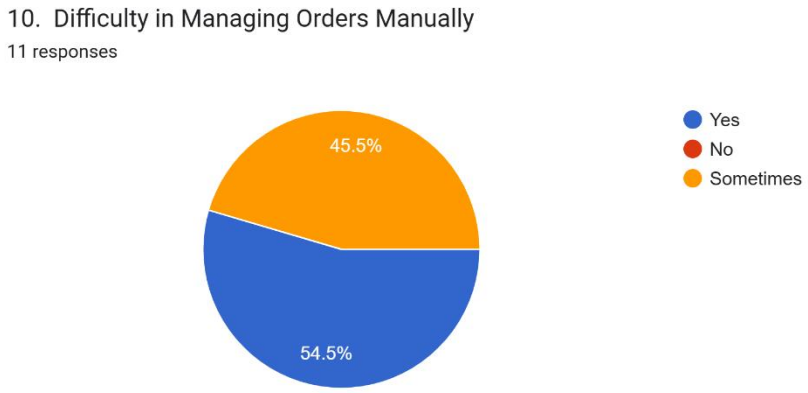

Figure 2.20: Survey Q10

A total of 11 responses were collected from participants regarding the difficulty of managing cake orders manually. The results indicate that 54.5% of respondents consistently experience difficulties, while 45.5% experience difficulties occasionally. Notably, no respondents indicated that manual order management is free of issues.

These findings clearly demonstrate that manual order handling processes are inefficient and prone to challenges such as errors, delays, and increased workload. The absence of negative responses (No) highlights the widespread need for improvement in current administrative workflows.

Overall, these results strongly justify the development of an automated order management system with features such as centralized order tracking, status updates, and administrative dashboards to reduce manual effort and improve operational efficiency.

Table 2.11: Survey Q10 Analysis

Response	Number of Responses	Percentage (%)
Yes	**6**	**54.5%**
Sometimes	**5**	**45.5%**
No	**0**	**0.0%**
Total	**11**	**100%**

Survey Question 11: Required Administrative Feature

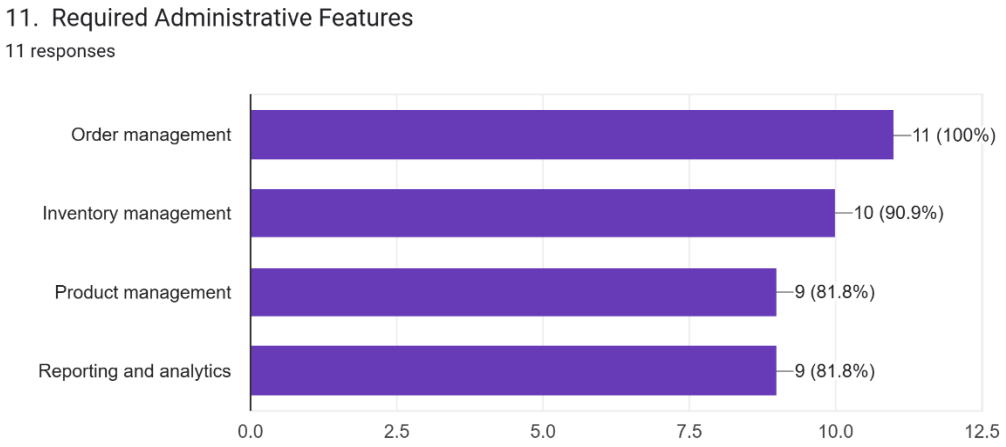

Figure 2.21: Survey Q11

A total of 11 responses were collected regarding required administrative features, with respondents allowed to select multiple options. The results indicate that order management is considered essential by 100% of respondents, highlighting its critical role in daily bakery operations.

Additionally, inventory management was selected by 90.9% of respondents, emphasizing the importance of tracking ingredient and product availability to prevent shortages and overproduction. Product management and reporting and analytics, each selected by 81.8% of respondents, further demonstrate the need for effective control over product offerings and access to business insights.

Table 2.12: Survey Q11 Analysis

Administrative Feature	Number of Responses	Percentage (%)
Order management	**11**	**100%**
Inventory management	**10**	**90.9%**
Product management	**9**	**81.8%**
Reporting and analytics	**9**	**81.8%**

Survey Question 12: Expected Benefit of an Automated System

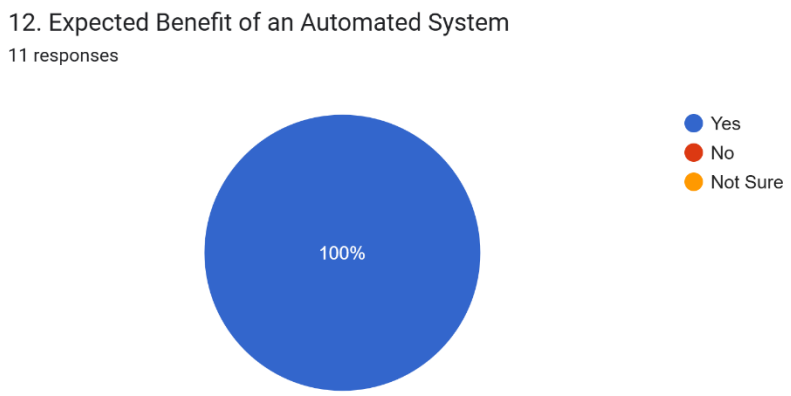

Figure 2.22: Survey Q12

A total of 11 responses were collected regarding the expected benefits of an automated system. The results show that 100% of respondents believe that an automated system would benefit bakery operations, indicating unanimous agreement on the need for digital transformation.

This strong consensus highlights the limitations of manual processes and reinforces the demand for automation to improve efficiency, accuracy, and overall workflow management. An automated system can reduce human errors, streamline order processing, and provide better visibility into operations through centralized management features.

Table 2.13: Survey Q12 Analysis

Response	Number of Responses	Percentage (%)
Yes	**11**	**100%**
No	**0**	**0.0%**
Not Sure	**0**	**0.0%**
Total	**11**	**100%**

3. Project Plan and Initial Design

This section presents the proposed methodology, development plan, and initial design approach for implementing the Digital Cake Ordering & Management System. It outlines how the system will be planned, developed, tested, and deployed to ensure that the identified functional and non-functional requirements are successfully achieved. In addition, this section introduces the preliminary system design concepts, including major system components and user interface considerations.

3.1 Components of the System

The Digital Cake Ordering & Management System is composed of three main components: software, hardware, and database. Each component plays a vital role in ensuring the smooth operation, reliability, and efficiency of the system. The following sections describe these components in detail.

3.1.1 Software Components

The software component includes all applications and platforms required to develop, operate, and manage the system. The proposed system is implemented as a web-based application, allowing users to access it through standard web browsers.

Application Software

- **Frontend Application:**
  The frontend provides the user interface for customers and administrators. Customers can browse cake categories, customize cakes, place orders, and track order status. Administrators can manage products, view orders, and update order progress. The interface is designed to be user-friendly, responsive, and accessible across different devices.

- **Backend Application:**
  The backend handles all business logic, including order processing, cake customization handling, payment processing, and user authentication. It acts as an intermediary between the frontend and the database, ensuring secure and accurate data transactions.

Supporting Software

- **Web Browser:**
  Users interact with the system through commonly available web browsers such as Google Chrome or Mozilla Firefox.
- **Server-side Software:**
  A web server and application runtime environment are required to host and execute the backend services.
- **Security Software:**
  Authentication and authorization mechanisms are implemented to protect user data and restrict access to administrative features.

3.1.2 Hardware Components

The hardware component consists of physical devices required to access, host, and maintain the system.

Client-side Hardware

- **User Devices:**
  Customers and administrators can access the system using devices such as desktop computers, laptops, tablets, or smartphones with internet connectivity.

Server-side Hardware

- **Application Server:**
  A server is required to host the web application and handle user requests.
- **Database Server:**
  A dedicated or shared server stores and manages the system database securely.

Networking Equipment

- **Internet Connectivity:**
  Reliable internet access is essential for real-time communication between users and the system.

- **Networking Devices:**
  Routers and firewalls are used to ensure secure and stable data transmission.

3.1.3 Database Components

The database component is responsible for storing, retrieving, and managing all system data. A RDBMS (relational database management system) is proposed to ensure data consistency and integrity.

Database Structure

The database stores information related to:

- **Users:** Customer and administrator details
- **Cakes:** Cake categories, product details, prices, and images
- **Orders:** Order details, customization options, order status, and timestamps
- **Payments:** Payment records and transaction references

Database Functions

- Ensures data accuracy and consistency
- Supports fast data retrieval for order processing and reporting
- Maintains data security through controlled access
- Enables order history and reporting for administrative purposes

3.2 Proposed Methodology

This section describes the proposed methodology adopted for the development of the Digital Cake Ordering & Management System. The methodology integrates the software, hardware, and database components discussed in the previous section to deliver a complete and reliable solution. A structured and iterative approach is followed to ensure that system requirements are accurately implemented and validated at each stage of development.

3.2.1 Methodology Overview

The development of the Digital Cake Ordering & Management System follows an iterative and incremental software development methodology. This methodology is selected because it allows continuous refinement of system features, early detection of errors, and frequent validation against user requirements. Instead of developing the entire system in a single phase, the system is built in manageable stages, ensuring flexibility and quality improvement throughout the development lifecycle.

3.2.2 Requirement Analysis and Planning

The first phase of the methodology involves a detailed requirement analysis, which forms the foundation of the entire system. Requirements were gathered and analyzed from multiple sources, including:

- The SRS (Software Requirements Specification) document
- Survey analysis conducted among online cake ordering users
- Evaluation of existing systems such as Kapruka, Taplow.lk, Dinevate, Winni, and Ferns N Petals

Functional requirements such as cake browsing, customization, order placement, online payment, order tracking, and administrative management were clearly identified. Non-functional requirements such as usability, security, performance, and scalability were also defined. Based on these requirements, a detailed project plan was prepared to guide the development process.

3.2.3 System Design Methodology

Once the requirements were finalized, the system design phase was initiated to determine how the solution would be structured and implemented.

Software Design Integration

The system is designed as a web-based application with a layered architecture:

- **Presentation Layer (Frontend):** Handles user interaction for customers and administrators.

- **Application Layer (Backend):** Implements business logic such as order processing, customization handling, and authentication.
- **Data Layer (Database):** Manages data storage and retrieval.

Database Design Approach

A relational database model is used to store structured data such as user details, cake information, orders, customization options, and payment records. Relationships between entities are defined to ensure data integrity and avoid redundancy.

Interface Design

The interface design of the Digital Cake Ordering & Management System focuses on providing a simple, intuitive, and user-friendly interaction for both customers and administrators. The interfaces are developed to support the core use cases defined in the Software Requirements Specification (SRS), ensuring smooth navigation, efficient task completion, and clear system feedback. Special emphasis is placed on usability, consistency, and accessibility to enhance the overall user experience and support effective system operation throughout the ordering and management processes.

In addition, the interfaces are designed to minimize user effort and reduce potential errors by incorporating clear visual feedback, logical workflow progression, and well-defined action buttons. Form inputs include guidance and validation messages to assist users in entering accurate information, while role-based access control ensures that users are presented only with functions relevant to their responsibilities. These design considerations improve system reliability, increase user satisfaction, and contribute to the effective adoption of the system by both customers and administrators.

Initial interfaces are prepared for major system interfaces, including:

- Home page
- Login page
- Order confirmation page
- Admin panel and Customer Panel
- Order tracking page

Home Page (Main Landing Interface)

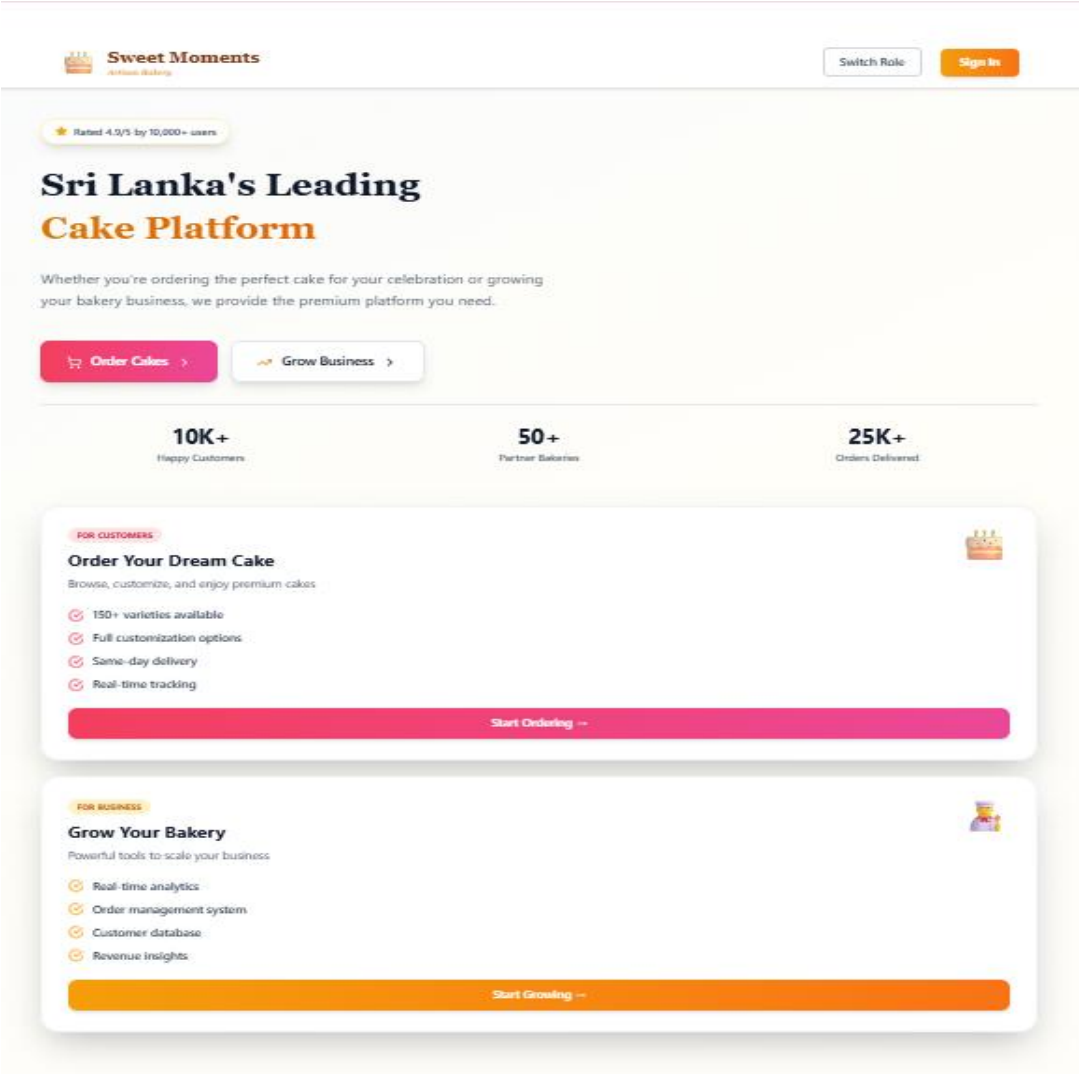

Figure 3.1: Home Page

The Home Page is the first interface of the Digital Cake Ordering & Management System and serves as the primary entry point for both customers and bakery owners. It introduces the platform *as* Sri Lanka's leading cake ordering solution and clearly highlights its dual purpose: enabling customers to order customized cakes and supporting bakeries in managing and growing their businesses.

The interface presents key navigation options such as Switch Role**,** Sign In, and call-to-action buttons including Order Cakes and Grow Business, allowing users to quickly proceed based on their role. For customers, the home page showcases features like cake variety availability, customization options, delivery services, featured cakes, and customer ratings, helping build trust and encourage engagement. For bakery owners, it

highlights business-oriented tools such as analytics dashboards, order management, customer insights, and revenue tracking.

Overall, the home page is designed to be visually appealing, informative, and user-friendly, guiding users seamlessly toward login, ordering, or business onboarding while clearly communicating the value and capabilities of the system.

Login Interface

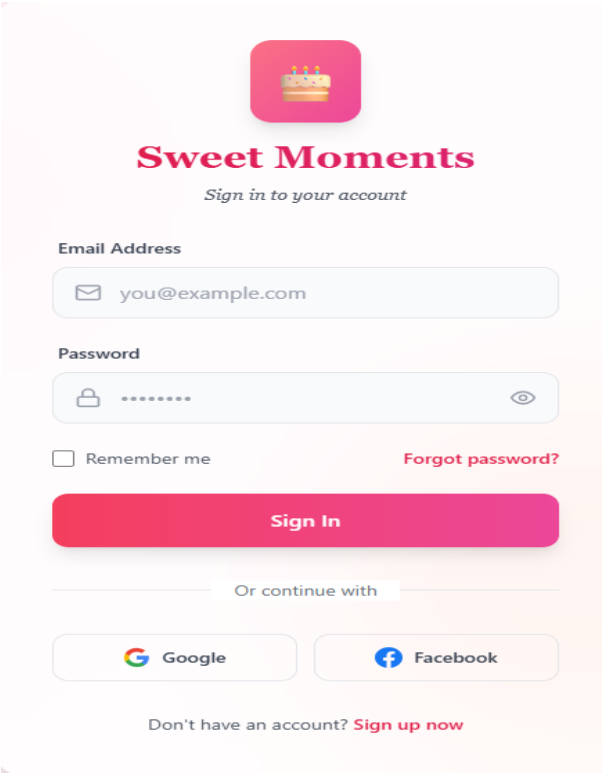

Figure 3.2: Login Interface

The Login interface of the Digital Cake Ordering & Management System enables users to securely sign in to their accounts using registered credentials. Users are required to enter their email address and password to access the system. The interface includes a "Remember me" option to simplify future logins and a "Forgot password?" link to assist users in recovering access to their accounts.

Additionally, the login screen supports alternative authentication methods such as Google and Facebook, providing convenient and faster sign-in options. Clear navigation is provided for new users through the "Sign up now" option, allowing easy account

creation. Overall, the login interface is designed to be simple, secure, and user-friendly, ensuring smooth access to both customer and administrator functionalities.

Admin Panel

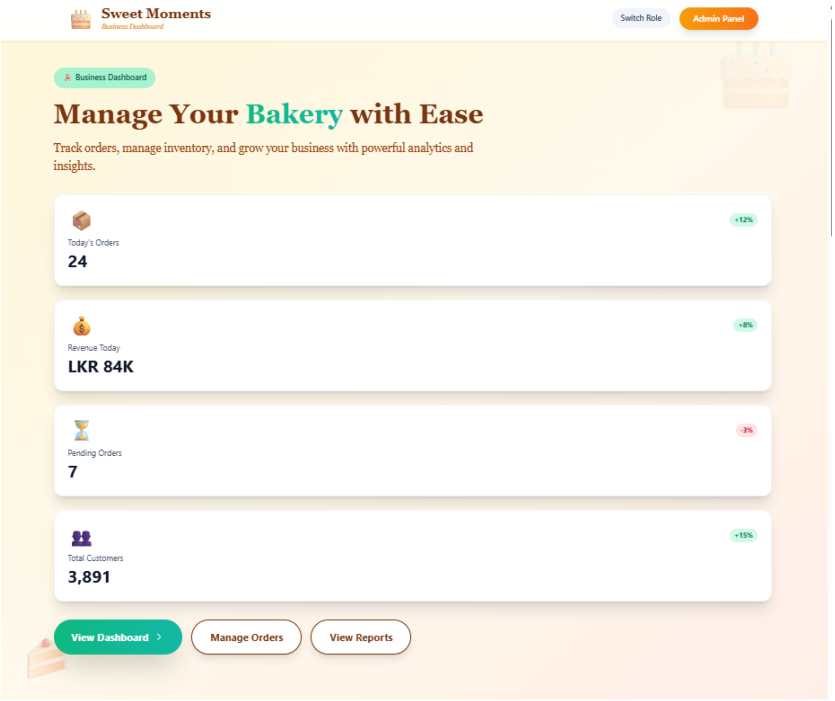

Figure 3.3: Admin Panel

The Admin Panel serves as the central control panel for bakery administrators in the Digital Cake Ordering & Management System. It provides a comprehensive overview of business performance, enabling administrators to efficiently manage daily operations. The admin panel displays key performance indicators such as today's orders, revenue, pending orders, and total customers, along with percentage changes to reflect business trends.

The interface includes quick navigation options such as View Dashboard, Manage Orders, and View Reports, allowing administrators to access essential management functions with ease. Designed for clarity and efficiency, the admin panel supports informed decision-making through real-time data insights, helping bakery owners track performance, manage orders effectively, and grow their business.

Customer Panel

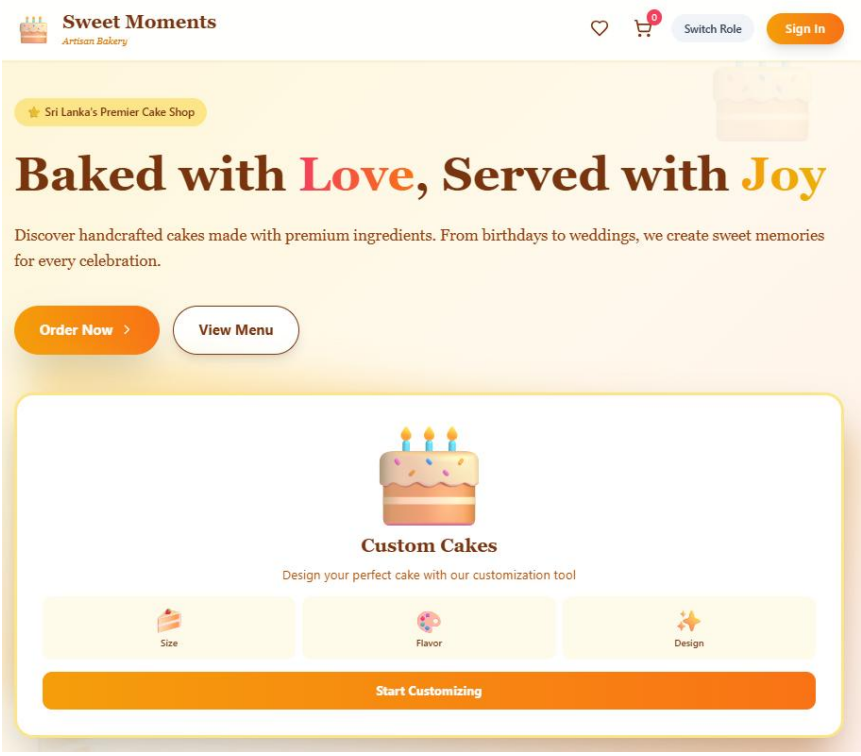

Figure 3.4: Customer Panel

The Customer Panel is the primary interface through which customers interact with the Digital Cake Ordering & Management System. It is designed to provide an engaging and intuitive experience for browsing, customizing, and ordering cakes. The interface highlights the bakery's offerings with clear navigation options such as Order Now, View Menu, and Start Customizing, enabling customers to easily begin their ordering journey.

The panel allows customers to explore cake options by occasion, view featured cakes with ratings and prices, and access customization features including cake size, flavor, and design. Informational sections emphasizing premium quality ingredients, fast delivery, and island-wide availability help build customer trust. Overall, the customer panel ensures a smooth, visually appealing, and user-friendly experience, guiding customers from product discovery to order initiation.

Order Processing Interface

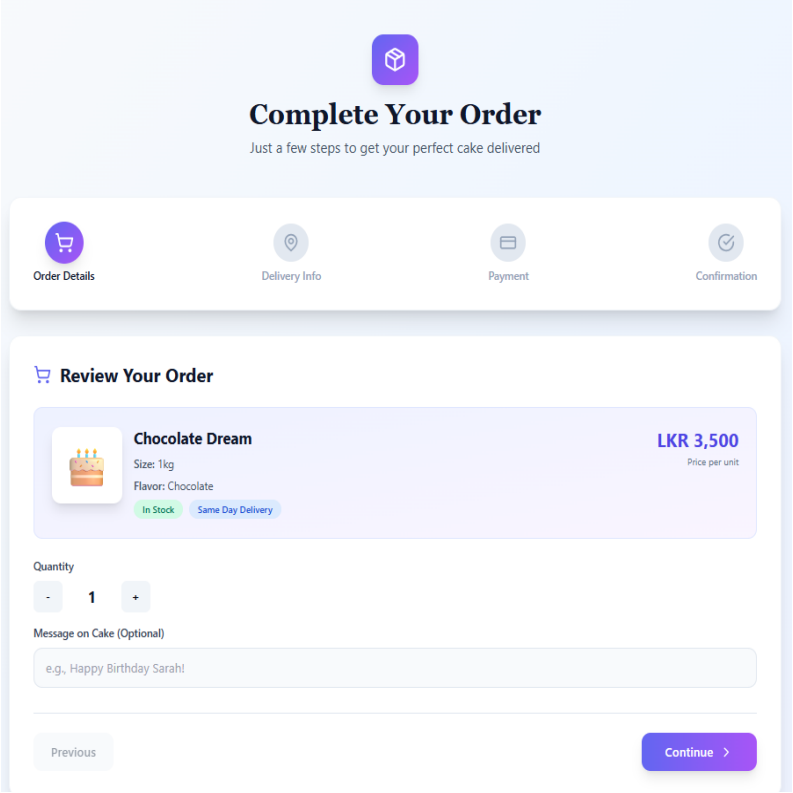

Figure 3.5: Order Processing Interface

The Order Processing Interface guides customers through the final steps of placing an order in the Digital Cake Ordering & Management System. It presents a clear, step-by-step flow including Order Details, Delivery Information, Payment, and Confirmation, helping customers understand their progress in the ordering process.

This interface allows customers to review selected cake details such as cake name, size, flavor, availability, delivery option, price, and quantity. Customers can also add a personalized message to be written on the cake. Navigation controls such as Previous and Continue enable easy movement between steps. The design ensures accuracy, transparency, and convenience, reducing errors and enhancing customer confidence before final order submission.

Order Tracking Interface

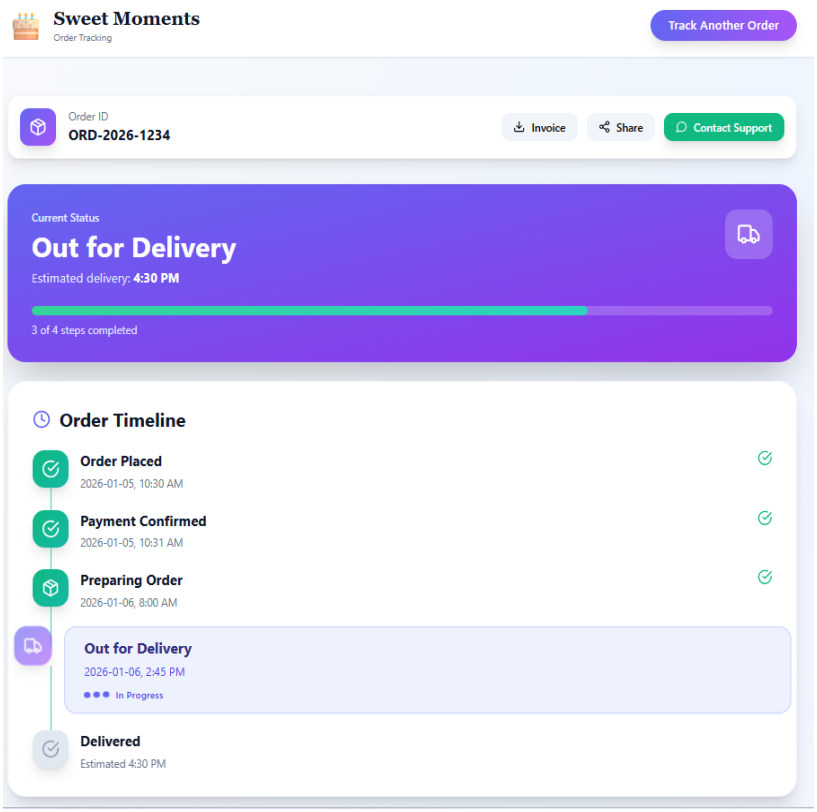

Figure 3.6: Order Tracking Interface

The Order Tracking Interface allows customers to monitor the real-time progress of their cake orders in the Digital Cake Ordering & Management System. Customers can track their order using a unique order ID and view the current order status along with an estimated delivery time. A visual progress indicator and detailed order timeline display each stage of the order lifecycle, from order placement and payment confirmation to preparation and delivery.

The interface also provides comprehensive order information, including delivery partner details, order items, delivery address, payment summary, and special instructions. Additional features such as invoice access, order sharing, contact support, and post-delivery rating enhance transparency and customer engagement. Overall, this interface improves customer confidence and satisfaction by providing clear, up-to-date order information and communication options throughout the delivery process.

3.2.4 System Implementation Strategy

During the ‘implementation phase, the system components are developed and integrated step by step.

- The frontend module is developed to provide a responsive and user-friendly interface accessible through web browsers on desktops and mobile devices.
- The backend module handles request processing, validation, order management, payment processing, and order status updates.
- The database module stores all transactional and reference data securely and supports fast data retrieval.

All modules are integrated with the necessary server-side hardware infrastructure to ensure reliable system operation.

3.2.5 Testing Methodology

Testing is a critical part of the proposed methodology and is conducted throughout the development lifecycle.

- **Unit Testing:** Individual modules such as order placement and customization are tested separately.
- **Integration Testing:** Ensures proper communication between frontend, backend, and database components.
- **System Testing:** Validates the complete system against functional and non-functional requirements.
- **User Acceptance Testing (UAT):** Confirms that the system meets user expectations and business needs.

Identified defects are resolved through repeated iterations, improving overall system quality.

3.2.6 Deployment Methodology

After successful testing, the system is deployed as a web-based application on a secure server environment. Deployment activities include:

- Configuring application and database servers
- Uploading and hosting system files
- Setting up security mechanisms
- Ensuring internet accessibility

Users can access the system using standard web browsers without installing additional software.

3.2.7 Maintenance and Continuous Improvement

Post-deployment, the system enters the maintenance phase. This includes:

- Monitoring system performance
- Fixing reported bugs
- Updating system features
- Enhancing security measures

User feedback collected after deployment is used to guide future system improvements.

3.3 Technologies Adapted

This section describes the technologies selected for the development, testing, data management, and deployment of the Digital Cake Ordering & Management System. The chosen technologies are based on system requirements, scalability, security, ease of maintenance, and compatibility with modern web standards. Each technology supports specific functional and non-functional requirements defined in the SRS.

3.3.1 Frontend Technologies

- **HTML5 (HyperText Markup Language, version 5)**
  Used to structure the web pages of the system. HTML5 enables semantic markup, improving accessibility and compatibility across browsers.
- **CSS3 (Cascading Style Sheets, version 3)**
  Responsible for styling and layout design. CSS3 is used to create a responsive

and visually appealing user interface suitable for both desktop and mobile devices.

- **JavaScript**
  Used to implement dynamic and interactive features such as form validation, dynamic cake customization options, and real-time UI (User Interface) updates.

- **React.js (JavaScript)**
  A modern JavaScript library used to build reusable UI components and manage application state efficiently. React enhances performance and improves user experience.

3.3.2 Backend Technologies

- **Node.js**
  Used as the server-side runtime environment to handle client requests efficiently. Node.js supports asynchronous processing, making it suitable for handling multiple user requests concurrently.

- **Express.js**
  A lightweight web application framework for Node.js used to develop RESTful APIs (Representational State Transfer Application Programming Interfaces), manage routing, and handle server-side logic such as order processing and authentication.

3.3.3 Database Technology

- **PostgreSQL**
  PostgreSQL is used as the primary relational database management system (RDBMS) for the Digital Cake Ordering & Management System. It is selected due to its robustness, high performance, strong data integrity, and support for complex queries.

PostgreSQL stores structured system data such as:

- User and administrator details
- Cake categories, product information, and pricing

- Order details and cake customization specifications
- Payment transaction records and order status history

Key advantages of using PostgreSQL include:

- Strong support for ACID (Atomicity, Consistency, Isolation, Durability) compliance, ensuring reliable transactions
- Advanced data integrity constraints such as foreign keys and checks
- High scalability and performance for multi-user systems
- Support for complex relationships required in order management systems
- Enhanced security through role-based access control

3.3.4 Testing Technologies

- **Manual Testing**
  Used to validate functional requirements such as cake ordering, customization, payment processing, and administrative operations.
- **Postman**
  Used to test backend APIs, verify request–response handling, and ensure correct integration between frontend and backend components.
- **Automated Testing (Optional – Jest/Mocha)**
  Used for unit and integration testing of backend services to improve reliability.

3.3.5 Deployment and Hosting Technologies

- **Web Server (Apache / Nginx)**
  Used to host the web application and manage HTTP (HyperText Transfer Protocol) requests.
- **Cloud Hosting Platform (e.g., AWS (Amazon Web Services) / Heroku / Vercel)**
  Used to deploy the application with scalability and high availability.

3.3.6 Development Tools

- **Visual Studio Code**
  Used as the primary development environment.
- **Git and GitHub**
  Used for version control, collaboration, and source code management.
- **Browser Developer Tools**
  Used for debugging and UI testing.

3.3.8 Security Technologies

- **Authentication and Authorization Mechanisms**
  Used to protect administrative features and user accounts.
- **HTTPS (HyperText Transfer Protocol Secure) and Data Encryption**
  Ensures secure communication between clients and the server.

3.4 Test and Deployment Plan

This section explains the testing strategies and deployment approach proposed for the Digital Cake Ordering & Management System. The testing plan focuses on validating system logic, graphical user interface (GUI), and overall functionality to ensure that the system meets the requirements defined in the SRS. In addition, this section describes the deployment process for the suggested organization and outlines the maintenance steps required to ensure long-term system reliability.

3.4.1 System Testing Plan

Testing is conducted throughout the system development lifecycle to identify errors, validate requirements, and ensure quality. The testing approach includes logical testing, GUI testing, and functional testing.

Logic Testing

Logic testing is performed to verify the correctness of the system's internal operations and business rules. This includes:

- Validation of order processing logic
- Verification of cake customization data handling

- Correct calculation of prices and totals
- Validation of order status transitions
- Verification of database interactions and constraints

Logic testing ensures that the system behaves as expected under different input conditions.

GUI Testing

GUI testing focuses on evaluating the usability, consistency, and visual correctness of the system interfaces. The following aspects are tested:

- Page layout and alignment across different devices
- Responsiveness of the interface on desktop and mobile screens
- Proper display of cake images, forms, and buttons
- Input validation messages and error handling
- Ease of navigation between system pages

GUI testing ensures a user-friendly experience for both customers and administrators.

Functional Testing

Functional testing is carried out to confirm that all system features operate according to the functional requirements specified in the SRS. This includes:

- User registration and login
- Cake browsing and customization
- Order placement and confirmation
- Online payment processing
- Order tracking for customers
- Order management and status updates by administrators

Each function is tested using predefined test cases to verify expected outputs.

Integration and User Acceptance Testing

- **Integration Testing:** Ensures seamless interaction between frontend, backend, and PostgreSQL database components.
- **User Acceptance Testing (UAT):** Conducted with selected users or bakery staff to validate system usability and requirement satisfaction.

3.4.2 Deployment Plan

The Digital Cake Ordering & Management System is deployed as a web-based application for use by the suggested organization, such as a bakery or cake shop.

Deployment Steps

1. **Server Preparation:**
   Configure the web server and database server environment.
2. **Database Setup:**
   Install PostgreSQL and create required tables, constraints, and roles.
3. **Application Deployment:**
   Upload frontend and backend application files to the hosting server.
4. **Configuration:**
   Configure environment variables, database connections, and security settings.
5. **Testing in Live Environment:**
   Perform final testing to ensure system stability.
6. **User Access Enablement:**
   Provide access to customers and administrators through web browsers.

3.4.3 Maintenance Plan

Post-deployment maintenance ensures continuous system operation and performance improvement.

Maintenance Activities

- Monitoring system performance and availability
- Fixing reported bugs and errors
- Applying security updates and patches
- Updating system features based on user feedback
- Performing regular database backups
- Managing user accounts and access control

Maintenance activities are conducted periodically to ensure system reliability and scalability.

4. References

[1] Kapruka, Cakes Online Sri Lanka, Kapruka.com, 2025. [Online]. Available: https://www.kapruka.com/lk/online/cakes. [Accessed: Jan. 2026].

[2] Taplow.lk, Cakes – Taplow Sri Lanka, Taplow.lk, 2025. [Online]. Available: https://taplow.lk/product-category/cakes/. [Accessed: Jan. 2026].

[3] Dinevate, *Bakery Website with Online Ordering*, Dinevate, 2024. [Online]. Available: https://www.dinevate.com/pages/bakery-website-with-ordering-sell-more-cakes-and-pastries-online. [Accessed: Jan. 2026].

[4] Winni, *Online Cake and Gift Delivery Platform*, Winni, 2024. [Online]. Available: https://www.winni.in/. [Accessed: Jan. 2026].

[5] Ferns N Petals, *Online Cake Ordering System*, Ferns N Petals, 2024. [Online]. Available: https://www.fnp.com/cakes. [Accessed: Jan. 2026].

[6] PostgreSQL Global Development Group, *PostgreSQL Documentation*, PostgreSQL, 2024. [Online]. Available: https://www.postgresql.org/docs/. [Accessed: Jan. 2026].

[7] Node.js Foundation, *Node.js Documentation*, OpenJS Foundation, 2024. [Online]. Available: https://nodejs.org/en/docs/. [Accessed: Jan. 2026].

[8] Express.js, *Express – Node.js Web Application Framework*, Open Source, 2024. [Online]. Available: https://expressjs.com/. [Accessed: Jan. 2026].

[9] React, *React Official Documentation*, Meta Platforms Inc., 2024. [Online]. Available: https://react.dev/. [Accessed: Jan. 2026].

[10] Google, *Google Forms*, Google LLC, 2024. [Online]. Available: https://www.google.com/forms/about/. [Accessed: Jan. 2026].

[11] I. Sommerville, *Software Engineering*, 10th ed. Boston, MA, USA: Pearson Education, 2016.

5. Appendix

This section contains supporting materials that complement the main content of the report. The appendix includes project planning details, requirement specifications, and system design diagrams that provide additional technical and visual information related to the Digital Cake Ordering & Management System.

5.1 Timeline – Gantt Chart

The Gantt chart below represents the planned schedule for the development of the Digital Cake Ordering & Management System. It illustrates the sequence, duration, and overlap of project activities from requirement analysis to deployment and documentation.

Project Activity	**Week 1**	**Week 2**	**Week 3**	**Week 4**	**Week 5**	**Week 6**
Requirement Analysis	■	■				
Literature Review & Survey	■	■				
System Design (UML & ERD)		■	■			
Database Design (PostgreSQL)		■				

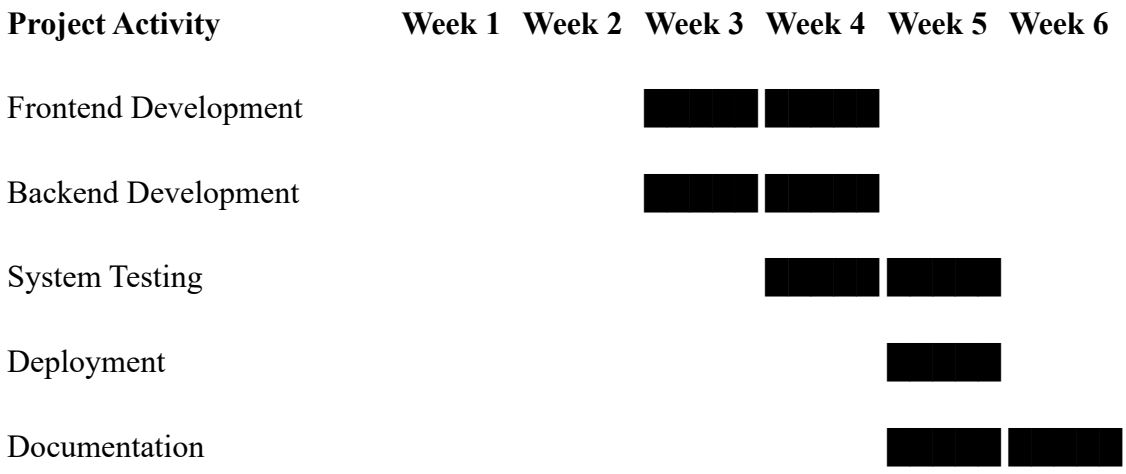

Figure 5.1: Gantt chart showing the project timeline of the Digital Cake Ordering & Management System

5.2 Software Requirements Specification (SRS)

This appendix includes the complete Software Requirements Specification (SRS) document prepared for the Digital Cake Ordering & Management System. The SRS defines the overall system scope, functional and non-functional requirements, assumptions, and constraints that guided the system design and implementation. It also specifies user roles, system features, data requirements, and interaction flows necessary to meet stakeholder expectations.

In addition, the SRS serves as a formal agreement between stakeholders and developers by clearly documenting system behavior and performance expectations. It provides a foundation for system analysis, interface design, database design, testing, and validation activities. The requirements documented in this appendix ensure traceability throughout the development lifecycle and support consistency between the proposed solution and the implemented system.

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5.3 Use Case Diagram

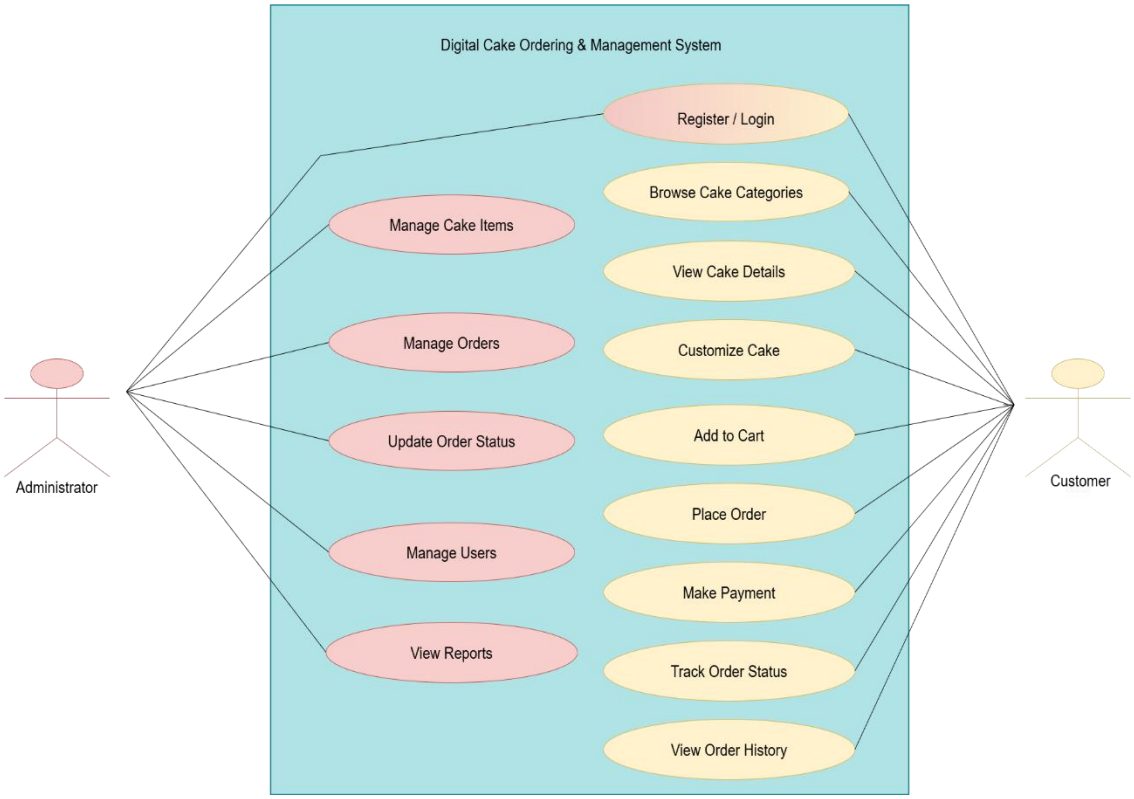

Figure 5.2: Use Case Diagram

The customer illustrates how customers interact with the Digital Cake Ordering & Management System to place and manage cake orders. Customers can register or log in to the system, browse cake categories, view cake details, and customize cakes according to their preferences. After selecting items, customers can add cakes to the cart, place orders, and complete payments through the system. Additionally, the diagram shows that customers can track order status and view their order history, enabling a convenient and efficient online cake ordering experience.

The administrator illustrates the functions performed by the administrator in managing the Digital Cake Ordering & Management System. The administrator can log in to the system and manage cake items by adding, updating, or removing product details. The diagram also shows the administrator's role in managing customer orders, updating order status, and handling user accounts. Additionally, the administrator can view reports related to orders and sales, supporting effective monitoring and operational control of the system.

5.4 Activity Diagrams

Activity Diagram 1: Cake Ordering Process

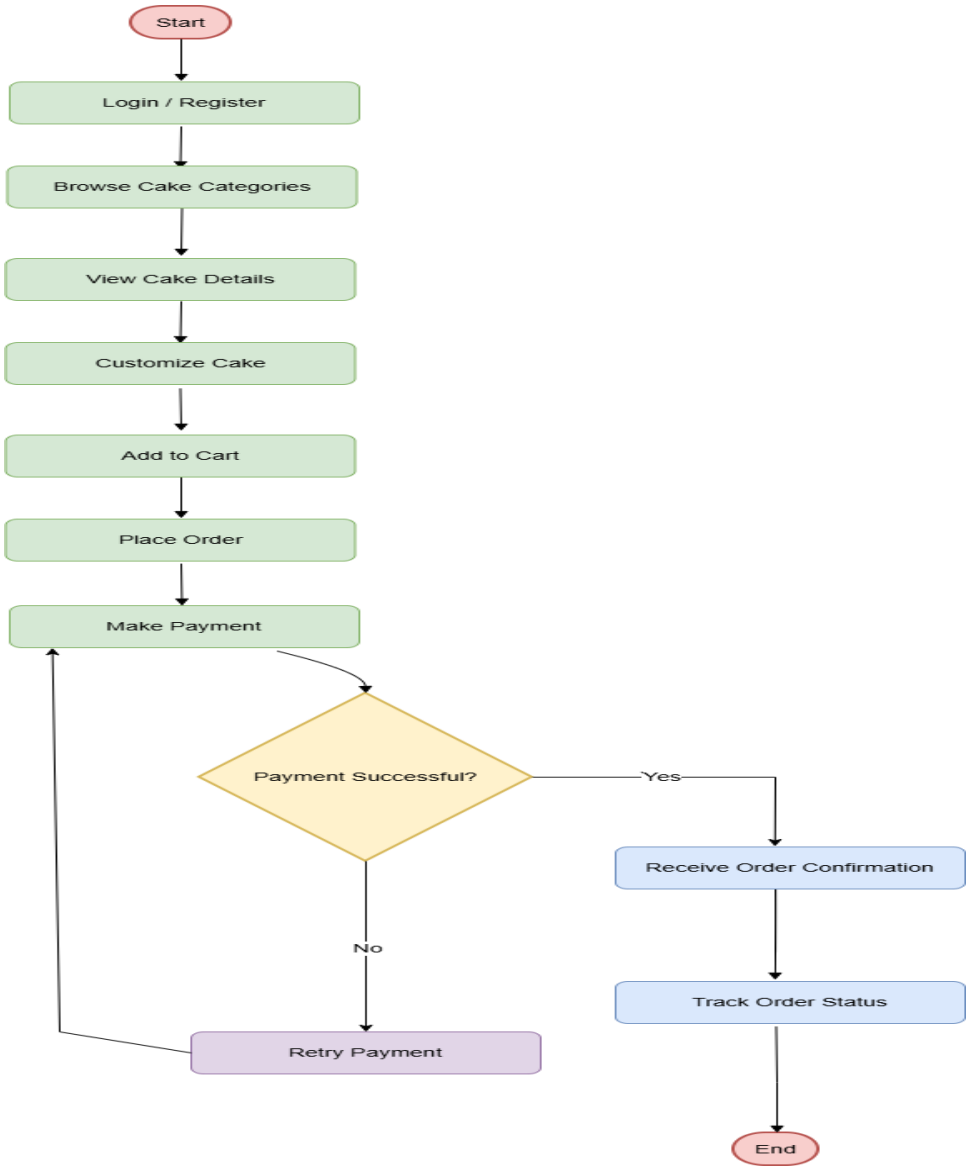

Figure 5.3: Activity Diagram 1: Cake Ordering Process

The activity diagram illustrates the workflow of the cake ordering process in the Digital Cake Ordering & Management System. It begins when a customer accesses the system and logs in. The customer then browses cake categories, views cake details, and customizes the selected cake according to preferences. After adding the cake to the cart, the customer places the order and completes the payment. Once payment is successful, the system confirms the order and updates the order status. The process ends when the customer receives the order confirmation.

Activity Diagram 2: Admin Order Management Process

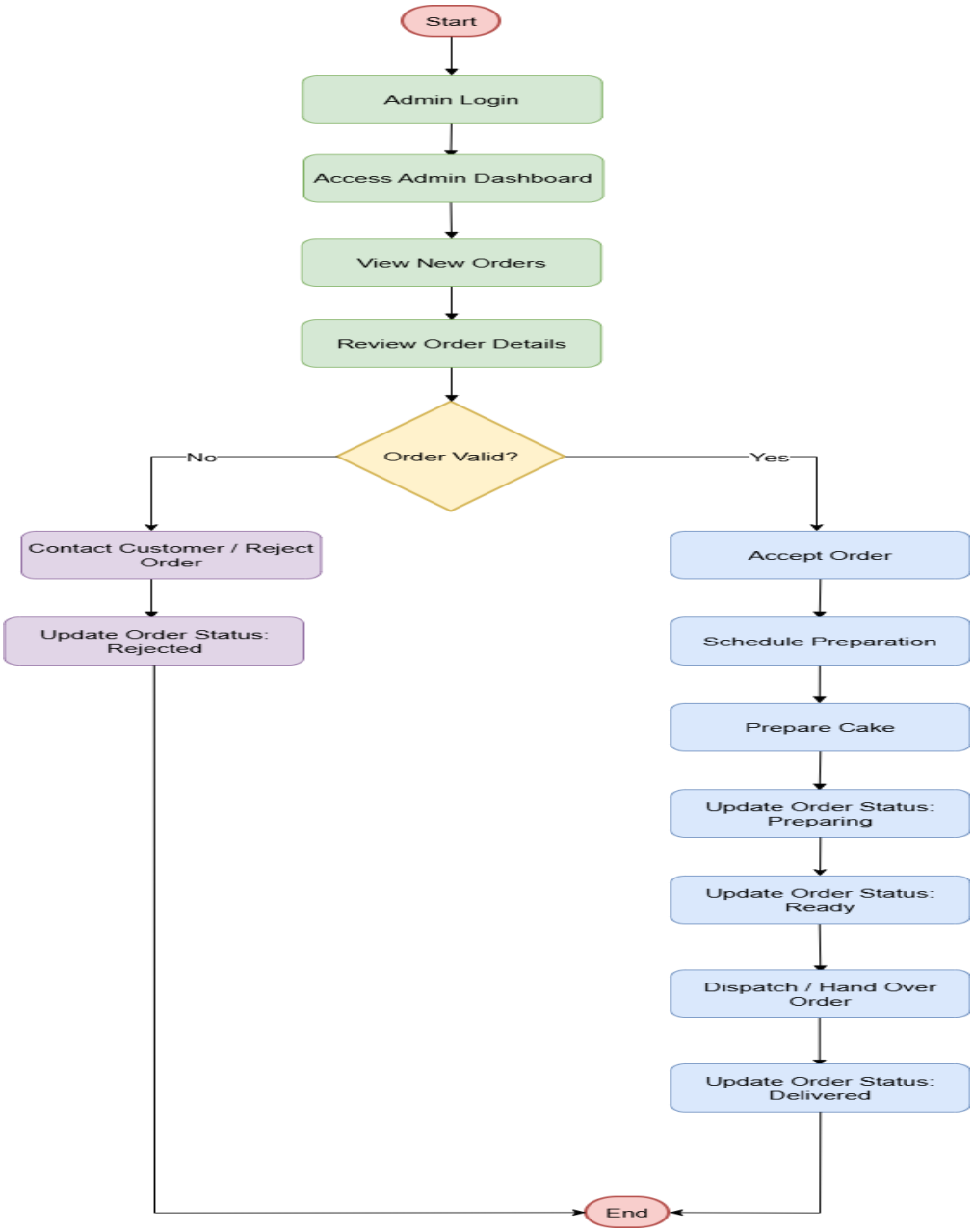

Figure 5.4: Activity Diagram 2: Admin Order Management Process

This activity diagram represents the workflow followed by the administrator in managing customer orders within the Digital Cake Ordering & Management System. It shows how the administrator logs into the system, reviews new orders, validates order details, prepares the cake, updates order status at each stage, and completes the order upon successful delivery.

5.5 Entity Relationship Diagram (ERD)

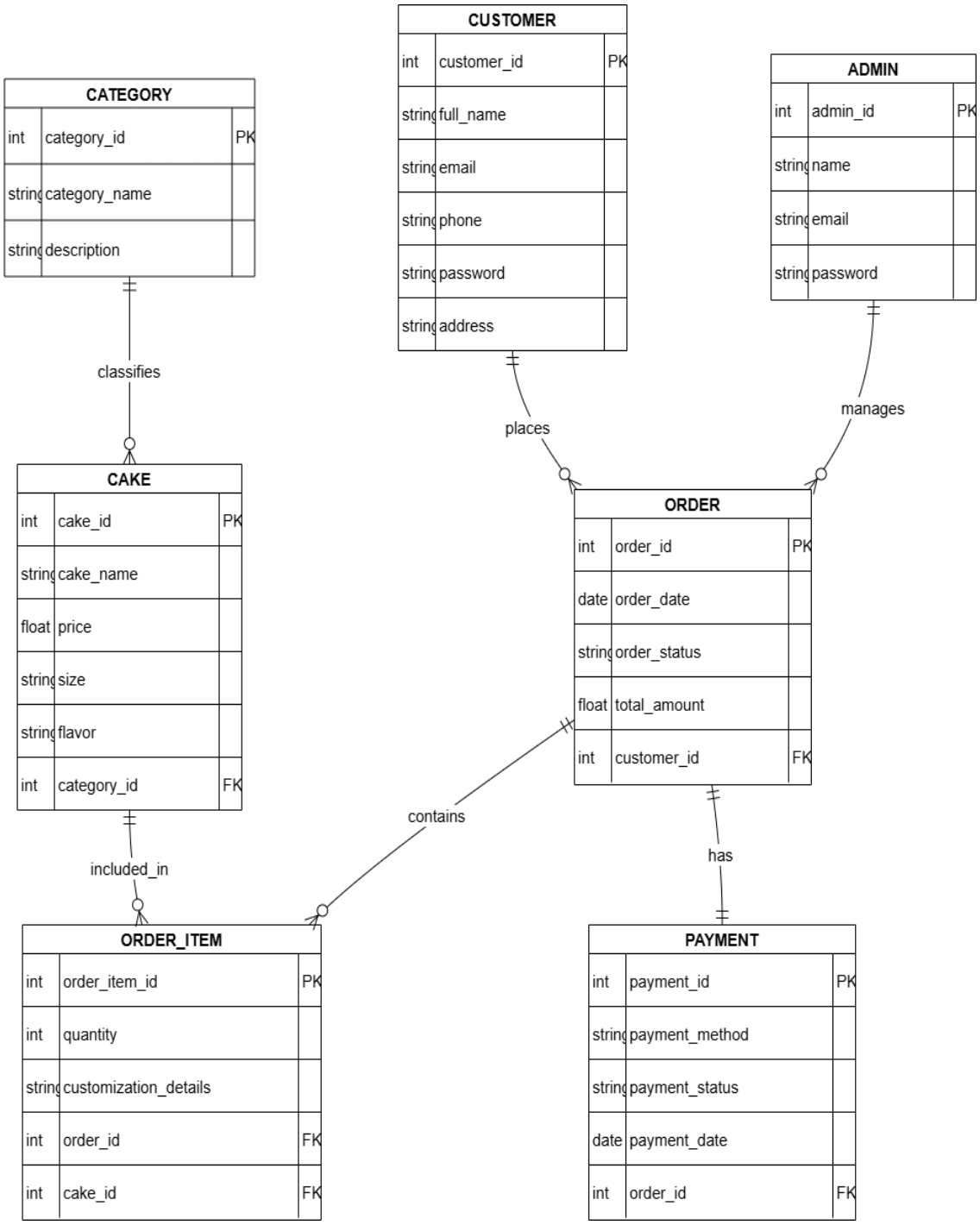

Figure 5.5: Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) represents the database structure of the Digital Cake Ordering & Management System by identifying key entities, their attributes, and the relationships among them. It shows how customers place orders, how orders consist

of customized cake items, and how payments are recorded for each order. The diagram also illustrates the role of the administrator in managing orders and the classification of cakes under different categories. This ERD supports efficient data management, ensures data consistency, and provides a solid foundation for implementing the system using a PostgreSQL database.